

Lecture 5 Image Denoising

(Deep Learning based methods)

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SIST Building-3 420

Outline

- Supervised Deep Learning Denoising
- Unsupervised Denoising
 - Deep Image Prior (DIP)
 - Noise2Noise: Learning Denoising from Noisy Data
 - Single Noisy Image: Blind-spot network (BSN), Noise2SR, Neighbor2Neighbor
 - Learning Image Distribution from Denoising (Denoising Diffusion Models)

Resource: <https://github.com/wenbihan/reproducible-image-denoising-state-of-the-art>

Supervised Deep Learning Approaches



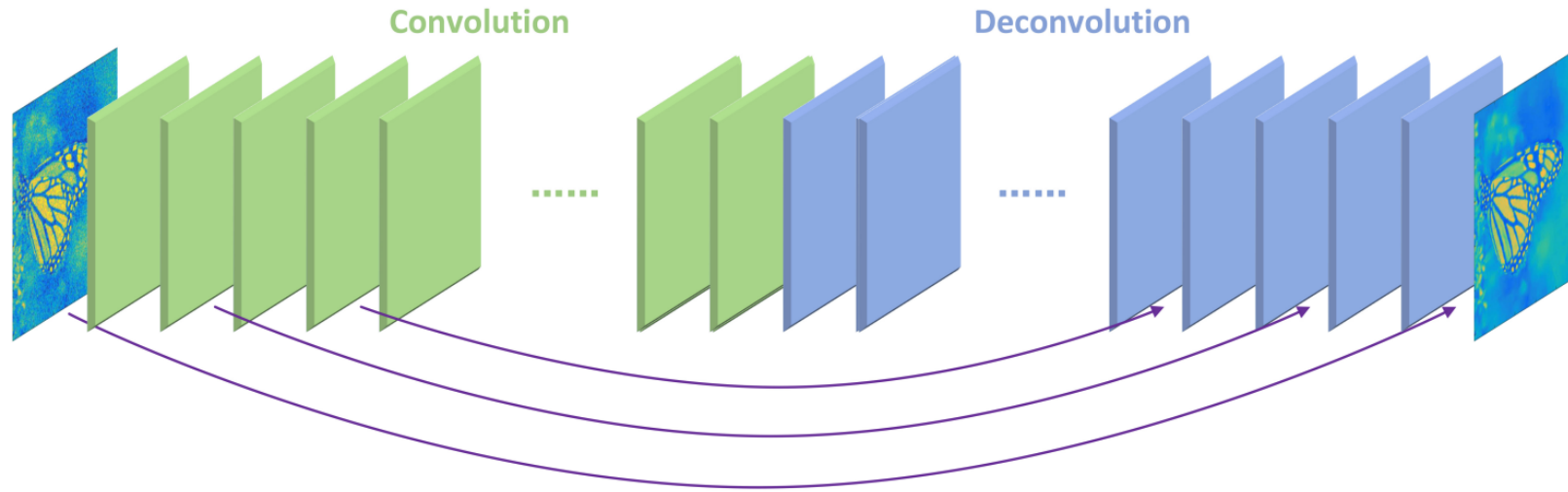
RED: Image Restoration Using Very Deep Convolutional Encoder-Decoder Networks with Symmetric Skip Connections

[1] Xiao-Jiao Mao , Chunhua Shen, Yu-Bin Yang: Image Restoration Using Very Deep Convolutional Encoder-Decoder Networks with Symmetric Skip Connections (NIPS2016), Mao et al. 2016

RED – Key Contribution

- The network is composed of multiple layers of convolution and de-convolution operators, learning end-to-end mappings from corrupted images to the original ones.
- We propose to symmetrically link convolutional and de-convolutional layers with skip-layer connections, with which the training converges much faster and attains a higher-quality local optimum.
- Significantly, with the large capacity, we can handle different levels of noises using a single model.

A deep noise to clean denoising method



We minimize the following Mean Squared Error(MSE):

$$\mathcal{L}(\Theta) = \frac{1}{n} \sum_{n=1}^N \|\mathcal{F}(X_i; \Theta) - Y_i\|_F^2. \quad (1)$$

Quantitative Evaluation Parameter

Table 1: Average PSNR and SSIM results of σ 10, 30, 50, 70 for the 14 images.

	PSNR								
	BM3D	EPLL	NCSR	PCLR	PGPD	WNNM	RED10	RED20	RED30
$\sigma = 10$	34.18	33.98	34.27	34.48	34.22	34.49	34.62	34.74	34.81
$\sigma = 30$	28.49	28.35	28.44	28.68	28.55	28.74	28.95	29.10	29.17
$\sigma = 50$	26.08	25.97	25.93	26.29	26.19	26.32	26.51	26.72	26.81
$\sigma = 70$	24.65	24.47	24.36	24.79	24.71	24.80	24.97	25.23	25.31
	SSIM								
$\sigma = 10$	0.9339	0.9332	0.9342	0.9366	0.9309	0.9363	0.9374	0.9392	0.9402
$\sigma = 30$	0.8204	0.8200	0.8203	0.8263	0.8199	0.8273	0.8327	0.8396	0.8423
$\sigma = 50$	0.7427	0.7354	0.7415	0.7538	0.7442	0.7517	0.7571	0.7689	0.7733
$\sigma = 70$	0.6882	0.6712	0.6871	0.6997	0.6913	0.6975	0.7012	0.7177	0.7206

Quantitative Evaluation Parameter

- PSNR (Peak signal to noise ratio)

$$MSE = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2$$

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right) = 20 \cdot \log_{10} \left(\frac{MAX_I}{\sqrt{MSE}} \right)$$

Quantitative Evaluation Parameter

- SSIM (Structure Similarity Index Measure)

- Luminance

$$\mu_x = \frac{1}{N} \sum_{i=1}^N x_i$$

$$l(x, y) = \frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1}$$

- Contrast

$$\sigma_x = \left(\frac{1}{N-1} \sum_{i=1}^N (x_i - \mu_x)^2 \right)^{\frac{1}{2}}$$

$$l(x, y) = \frac{2\sigma_x\sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2}$$

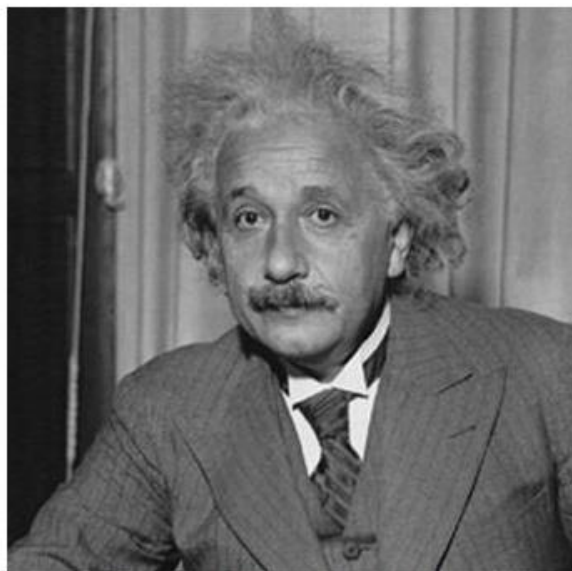
- Structure

$$c(x, y) = \frac{2\sigma_{xy} + C_3}{\sigma_x\sigma_y + C_3}$$

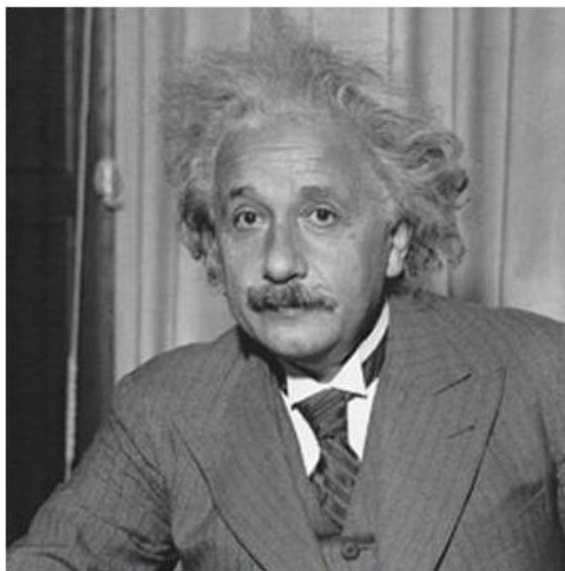
$$\sigma_{xy} = \frac{1}{N-1} \sum_{i=1}^N (x_i - \mu_x)(y_i - \mu_y)$$

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

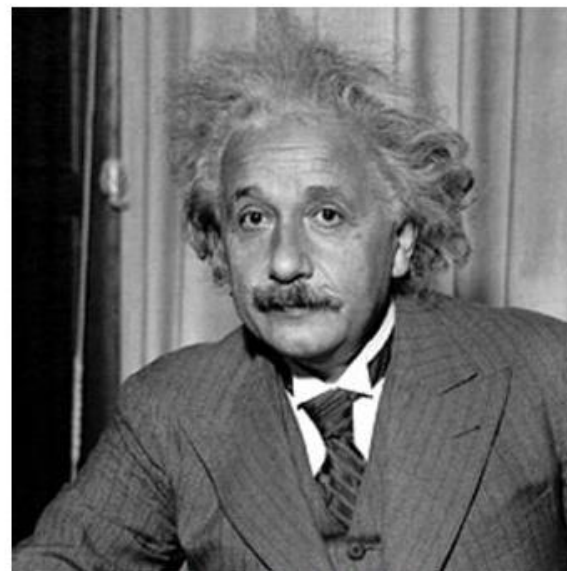




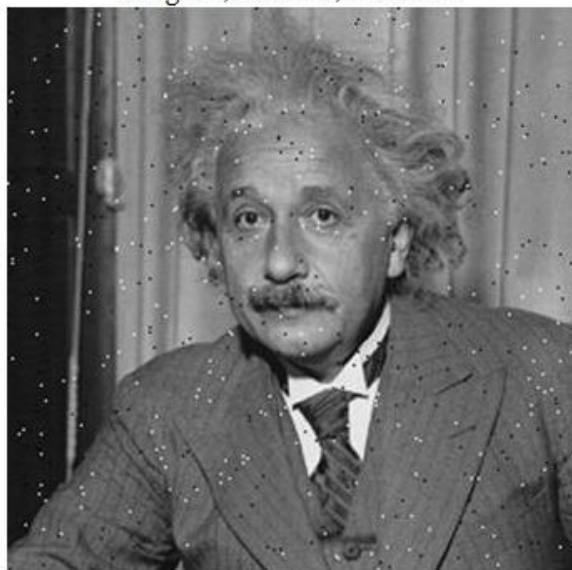
Original, MSE = 0; SSIM = 1



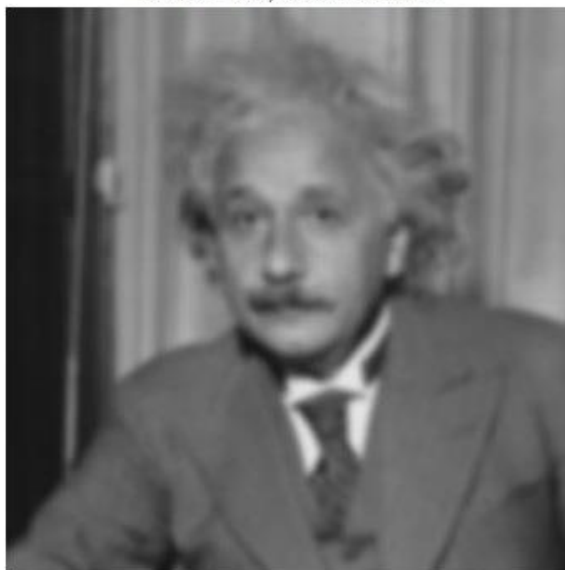
MSE = 144, SSIM = 0.988



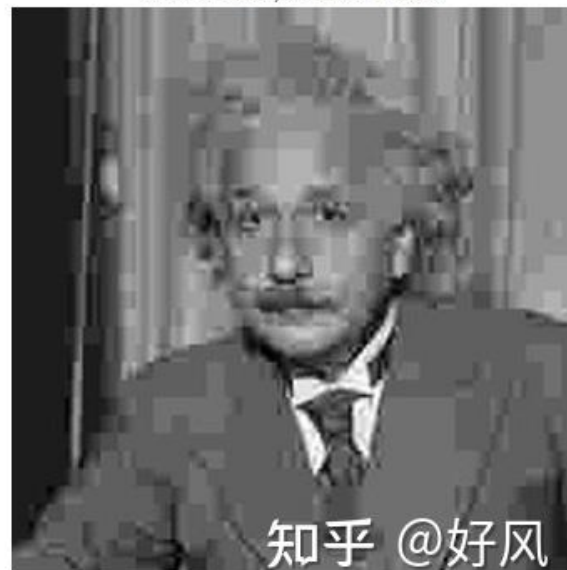
MSE = 144, SSIM = 0.913



MSE = 144, SSIM = 0.840



MSE = 144, SSIM = 0.694



MSE = 142, SSIM = 0.662

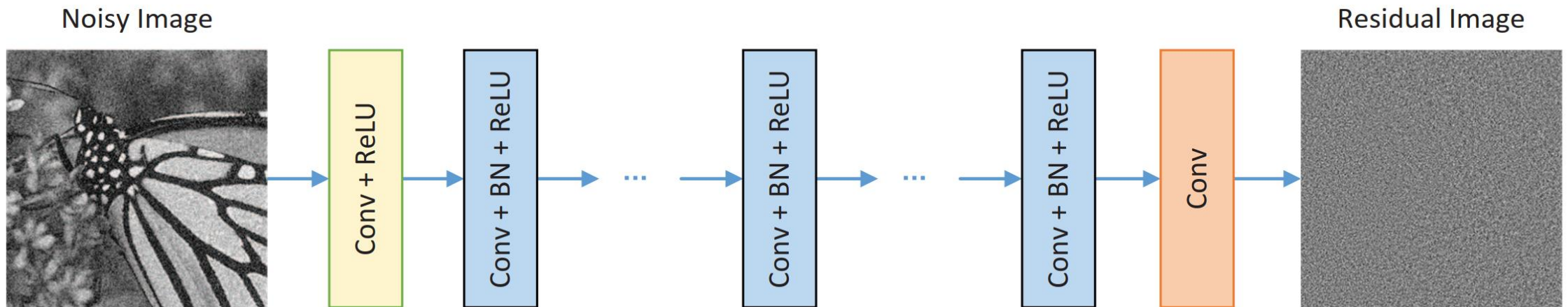
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DnCNN: Beyond a Gaussian Denoiser: Residual Learning of Deep CNN for Image Denoising

[1] Kai Zhang, Wangmeng Zuo, [Lei Zhang](#) FFDNet: Toward a Fast and Flexible Solution for CNN-Based Image Denoising (TIP 2018), et al.

DnCNN– Key Contribution

- With the residual learning strategy, DnCNN implicitly removes the latent clean image in the hidden layers.
- Specifically, residual learning and batch normalization are utilized to speed up the training process as well as boost the denoising performance.



Deep Residual Learning for Image Recognition

Deep Residual Learning for Image Recognition

- Motivation

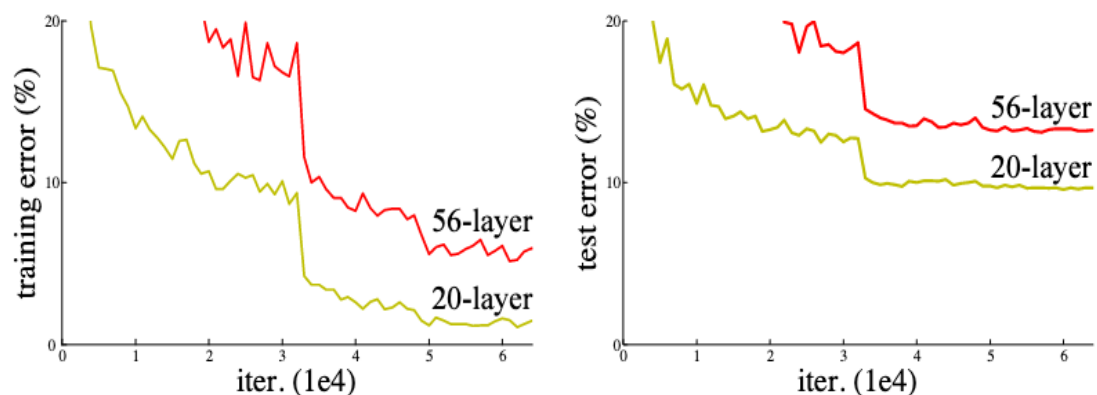


Figure 1. Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer “plain” networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

- Structure

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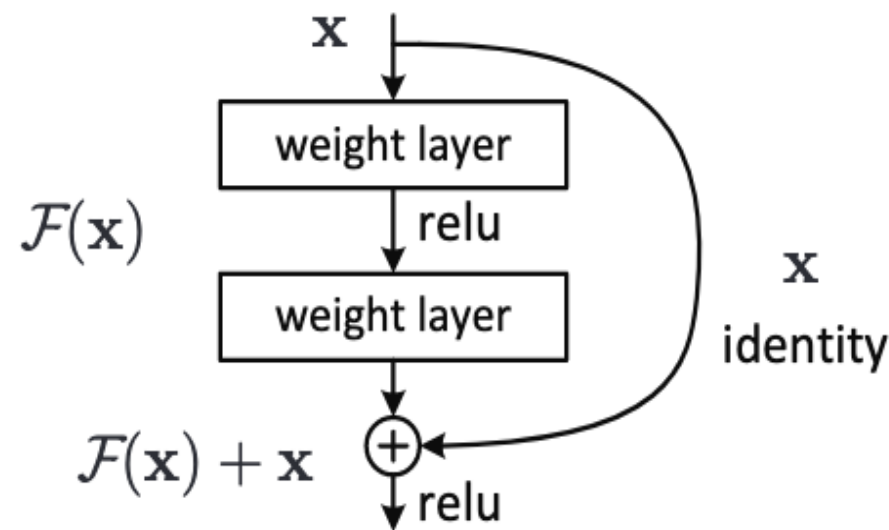


Figure 2. Residual learning: a building block.

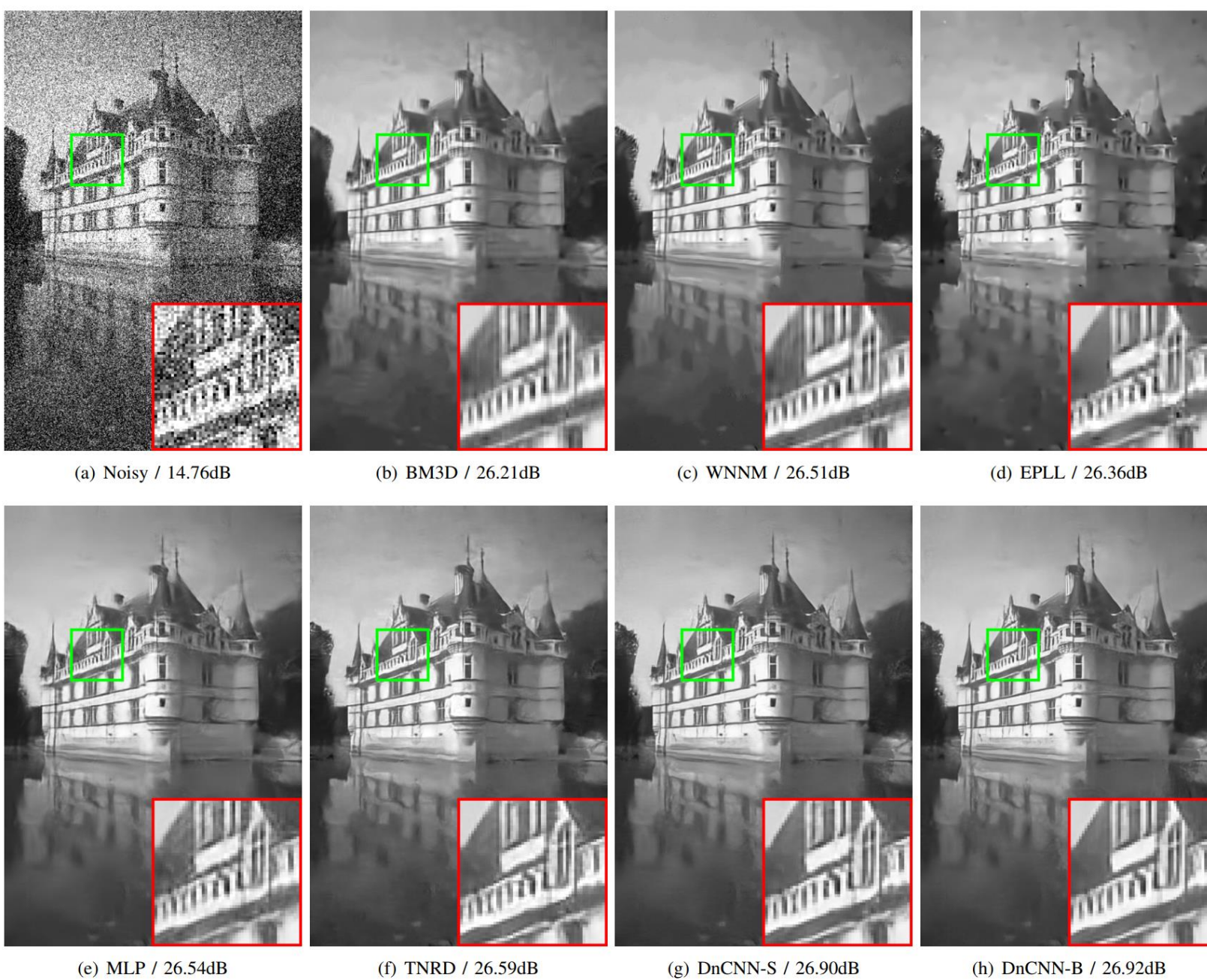


Fig. 4. Denoising results of one image from BSD68 with noise level 50.

FFDNet: Toward a Fast and Flexible Solution for CNN based Image Denoising

[1] FFDNet: Toward a Fast and Flexible Solution for CNN-Based Image Denoising (TIP 2018), Zhang et al.

FFDNet – Key Contribution

- A fast and flexible denoising network, namely FFDNet, is proposed for discriminative image denoising. By taking a tunable noise level map as input, a single FFDNet is able to deal with noise on different levels, as well as spatially variant noise.
- We highlight the importance to guarantee the role of the noise level map in controlling the trade-off between noise reduction and detail preservation.
- FFDNet exhibits perceptually appealing results on both synthetic noisy images corrupted by AWGN and real world noisy images, demonstrating its potential for practical image denoising.

Pipeline

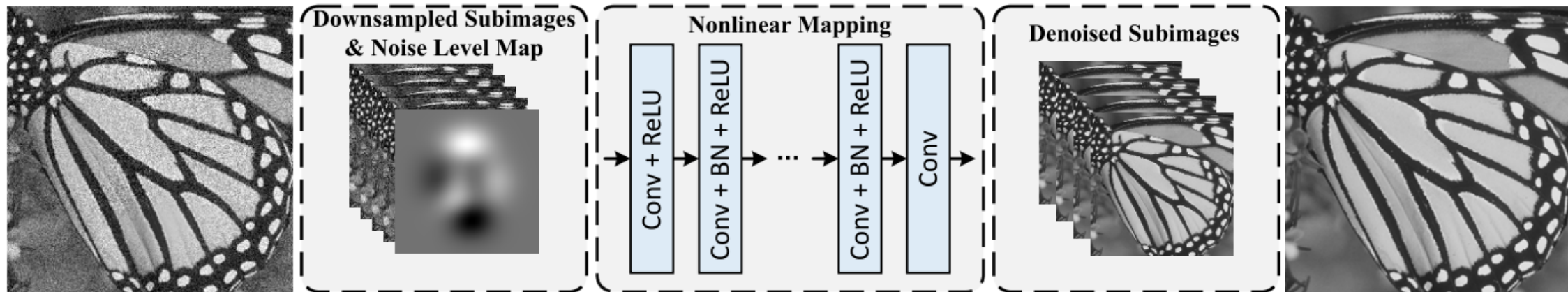


Fig. 1. The architecture of the proposed FFDNet for image denoising. The input image is reshaped to four sub-images, which are then input to the CNN together with a noise level map. The final output is reconstructed by the four denoised sub-images.

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{x}\|^2 + \lambda \Phi(\mathbf{x}), \quad (1)$$

where $\frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{x}\|^2$ is the data fidelity term with noise level σ , $\Phi(\mathbf{x})$ is the regularization term associated with image prior, and λ controls the balance between the data fidelity and regularization terms. It is worth noting that in practice λ governs the compromise between noise reduction and detail preservation. When it is too small, much noise will remain; on the opposite, details will be smoothed out along with suppressing noise.

With some optimization algorithms, the solution of Eqn. (1) actually defines an implicit function given by

$$\hat{\mathbf{x}} = \mathcal{F}(\mathbf{y}, \sigma, \lambda; \Theta). \quad (2)$$

Since λ can be absorbed into σ , Eqn. (2) can be rewritten as

$$\hat{\mathbf{x}} = \mathcal{F}(\mathbf{y}, \sigma; \Theta). \quad (3)$$



Pipeline

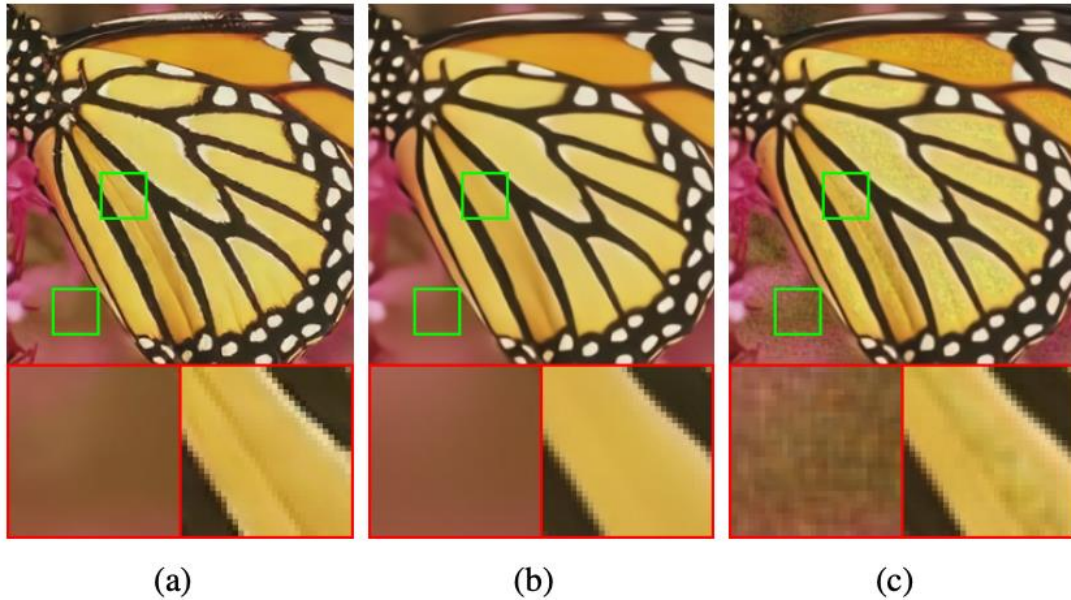
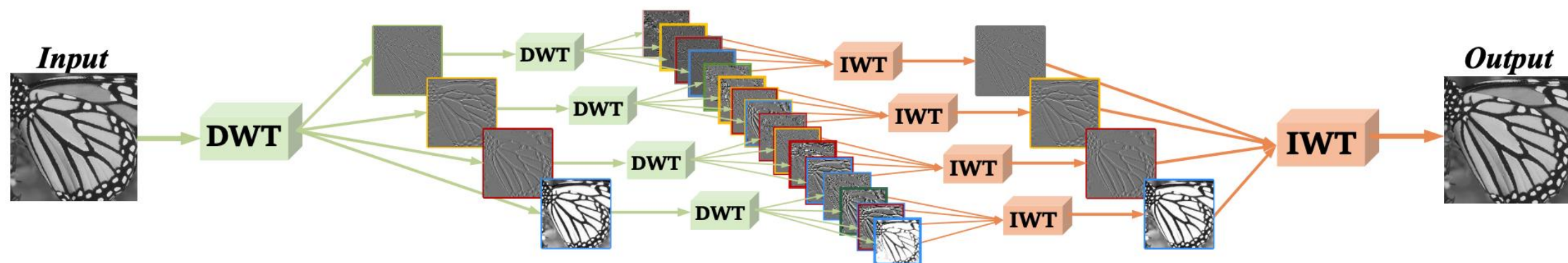


Fig. 2. An example to show the importance of guaranteeing the role of noise level map in controlling the trade-off between noise reduction and detail preservation. The input is a noisy image with noise level 25. (a) Result without visual artifacts by matched noise level 25. (b) Result without visual artifacts by mismatched noise level 60. (c) Result with visual artifacts by mismatched noise level 60.

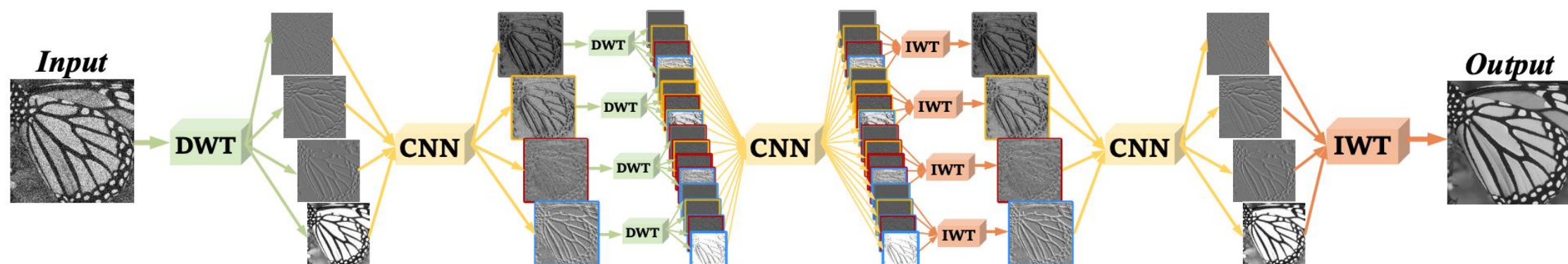
Images	C.man	House	Peppers	Starfish	Monarch	Airplane	Parrot	Lena	Barbara	Boat	Man	Couple	Average
Noise Level	$\sigma = 15$												
BM3D	31.91	34.93	32.69	31.14	31.85	31.07	31.37	34.26	33.10	32.13	31.92	32.10	32.37
WNNM	32.17	35.13	32.99	31.82	32.71	31.39	31.62	34.27	33.60	32.27	32.11	32.17	32.70
TNRD	32.19	34.53	33.04	31.75	32.56	31.46	31.63	34.24	32.13	32.14	32.23	32.11	32.50
DnCNN	32.61	34.97	33.30	32.20	33.09	31.70	31.83	34.62	32.64	32.42	32.46	32.47	32.86
FFDNet	32.42	35.01	33.10	32.02	32.77	31.58	31.77	34.63	32.50	32.35	32.40	32.45	32.75
Noise Level	$\sigma = 25$												
BM3D	29.45	32.85	30.16	28.56	29.25	28.42	28.93	32.07	30.71	29.90	29.61	29.71	29.97
WNNM	29.64	33.22	30.42	29.03	29.84	28.69	29.15	32.24	31.24	30.03	29.76	29.82	30.26
MLP	29.61	32.56	30.30	28.82	29.61	28.82	29.25	32.25	29.54	29.97	29.88	29.73	30.03
TNRD	29.72	32.53	30.57	29.02	29.85	28.88	29.18	32.00	29.41	29.91	29.87	29.71	30.06
DnCNN	30.18	33.06	30.87	29.41	30.28	29.13	29.43	32.44	30.00	30.21	30.10	30.12	30.43
FFDNet	30.06	33.27	30.79	29.33	30.14	29.05	29.43	32.59	29.98	30.23	30.10	30.18	30.43
Noise Level	$\sigma = 35$												
BM3D	27.92	31.36	28.51	26.86	27.58	26.83	27.40	30.56	28.98	28.43	28.22	28.15	28.40
WNNM	28.08	31.92	28.75	27.27	28.13	27.10	27.69	30.73	29.48	28.54	28.33	28.24	28.69
MLP	28.08	31.18	28.54	27.12	27.97	27.22	27.72	30.82	27.62	28.53	28.47	28.24	28.46
DnCNN	28.61	31.61	29.14	27.53	28.51	27.52	27.94	30.91	28.09	28.72	28.66	28.52	28.82
FFDNet	28.54	31.99	29.18	27.58	28.54	27.47	28.02	31.20	28.29	28.82	28.70	28.68	28.92
Noise Level	$\sigma = 50$												
BM3D	26.13	29.69	26.68	25.04	25.82	25.10	25.90	29.05	27.22	26.78	26.81	26.46	26.72
WNNM	26.45	30.33	26.95	25.44	26.32	25.42	26.14	29.25	27.79	26.97	26.94	26.64	27.05
MLP	26.37	29.64	26.68	25.43	26.26	25.56	26.12	29.32	25.24	27.03	27.06	26.67	26.78
TNRD	26.62	29.48	27.10	25.42	26.31	25.59	26.16	28.93	25.70	26.94	26.98	26.50	26.81
DnCNN	27.03	30.00	27.32	25.70	26.78	25.87	26.48	29.39	26.22	27.20	27.24	26.90	27.18
FFDNet	27.03	30.43	27.43	25.77	26.88	25.90	26.58	29.68	26.48	27.32	27.30	27.07	27.32
Noise Level	$\sigma = 75$												
BM3D	24.32	27.51	24.73	23.27	23.91	23.48	24.18	27.25	25.12	25.12	25.32	24.70	24.91
WNNM	24.60	28.24	24.96	23.49	24.31	23.74	24.43	27.54	25.81	25.29	25.42	24.86	25.23
MLP	24.63	27.78	24.88	23.57	24.40	23.87	24.55	27.68	23.39	25.44	25.59	25.02	25.07
DnCNN	25.07	27.85	25.17	23.64	24.71	24.03	24.71	27.54	23.63	25.47	25.64	24.97	25.20
FFDNet	25.29	28.43	25.39	23.82	24.99	24.18	24.94	27.97	24.24	25.64	25.75	25.29	25.49

Multi-level Wavelet-CNN for Image Restoration

[1] Multi-level Wavelet-CNN for Image Restoration (CVPR 2018), Liu et al.



(a) Multi-level WPT architecture



(b) Multi-level wavelet-CNN architecture

Figure 2. From WPT to MWCNN. Intuitively, WPT can be seen as a special case of our MWCNN without CNN blocks.

Key Contribution

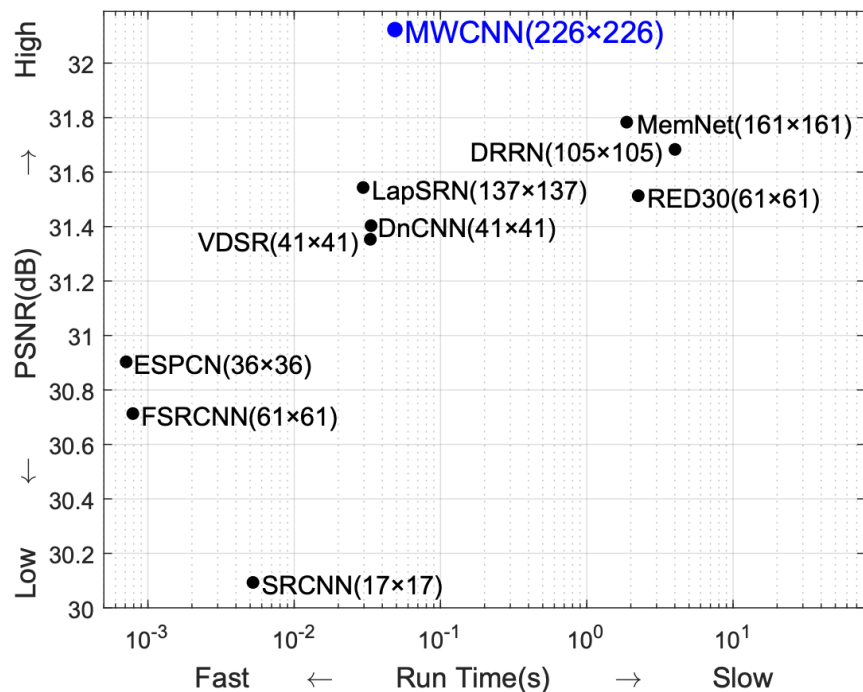


Figure 1. The run time vs. PSNR value of representative CNN models, including SRCNN [16], FSRCNN [14], ESPCN [45], VDSR [29], DnCNN [57], RED30 [37], LapSRN [31], DRRN [47], MemNet [47] and our MWCNN. The receptive field of each model are also provided. The PSNR and time are evaluated on Set5 with the scale factor $\times 4$ running on a GTX1080 GPU.

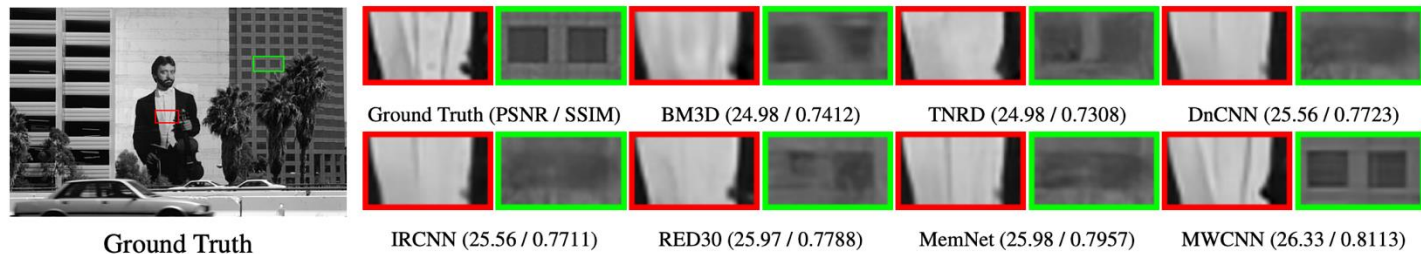


Figure 5. Image denoising results of "Test011" (BSD68) with noise level 50.



Figure 6. Single image super-resolution results of "barbara" (Set14) with upscaling factor $\times 4$.

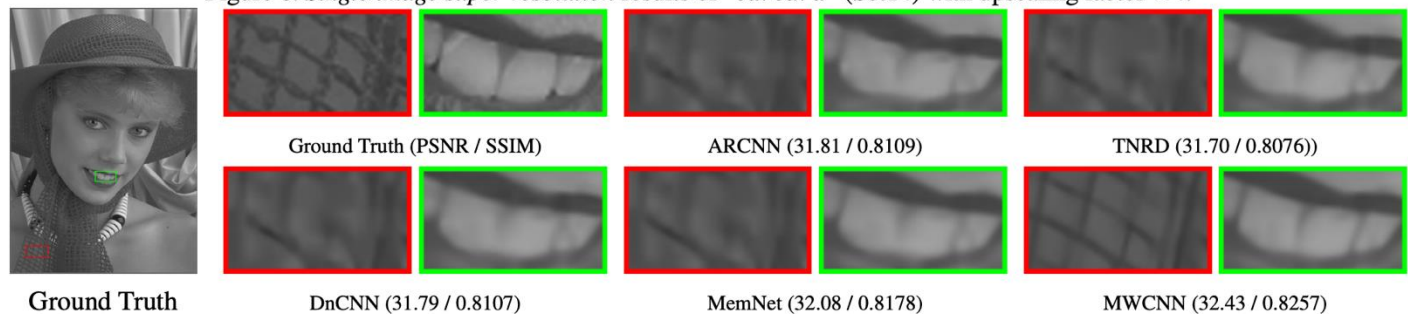


Figure 7. JPEG image artifacts removal results of "womanhat" (LIVE1) with quality factor 10.

Image Denoising					
Size	TNRD [10]	DnCNN [57]	RED30 [37]	MemNet [47]	MWCNN
256×256	0.010	0.0143	1.362	0.8775	0.0586
512×512	0.032	0.0487	4.702	3.606	0.0907
1024×1024	0.116	0.1688	15.77	14.69	0.3575
Single Image Super-Resolution					
Size	VDSR [29]	LapSRN [31]	DRRN [47]	MemNet [37]	MWCNN
256×256	0.0172	0.0229	3.063	0.8774	0.0424
512×512	0.0575	0.0357	8.050	3.605	0.0780
1024×1024	0.2126	0.1411	25.23	14.69	0.3167
JPEG Image Artifacts Removal					
Size	ARCNN [15]	TNRD [10]	DnCNN [57]	MemNet [37]	MWCNN
256×256	0.0277	0.009	0.0157	0.8775	0.0531
512×512	0.0532	0.028	0.0568	3.607	0.0811
1024×1024	0.1613	0.095	0.2012	14.69	0.2931

DIP: Deep Image Prior

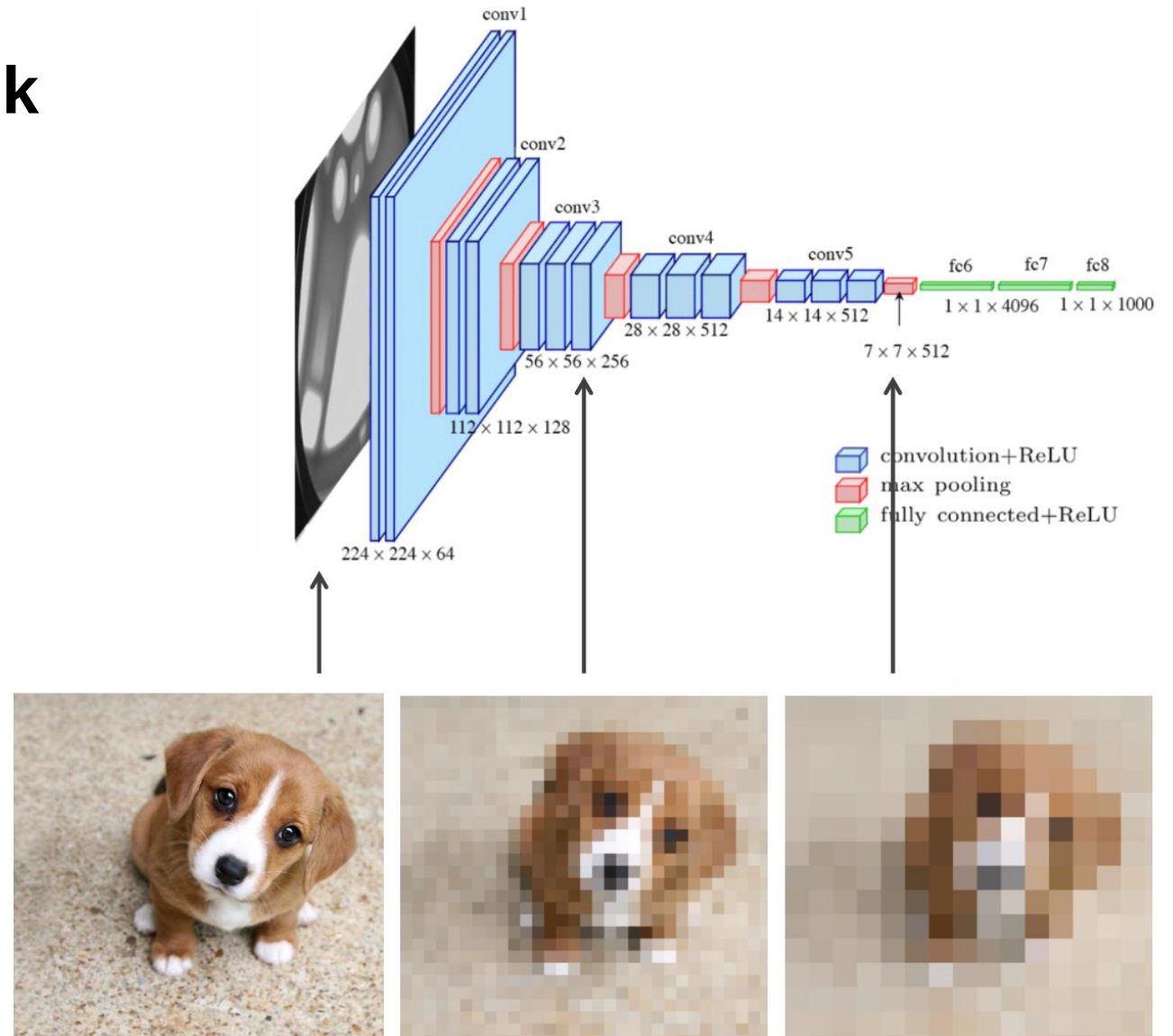
[1] Deep Image Prior (CVPR 2018), Ulyanov et al.



Digital Image Prior: CNN for Denoising

Convolutional Neural Network

- VGG



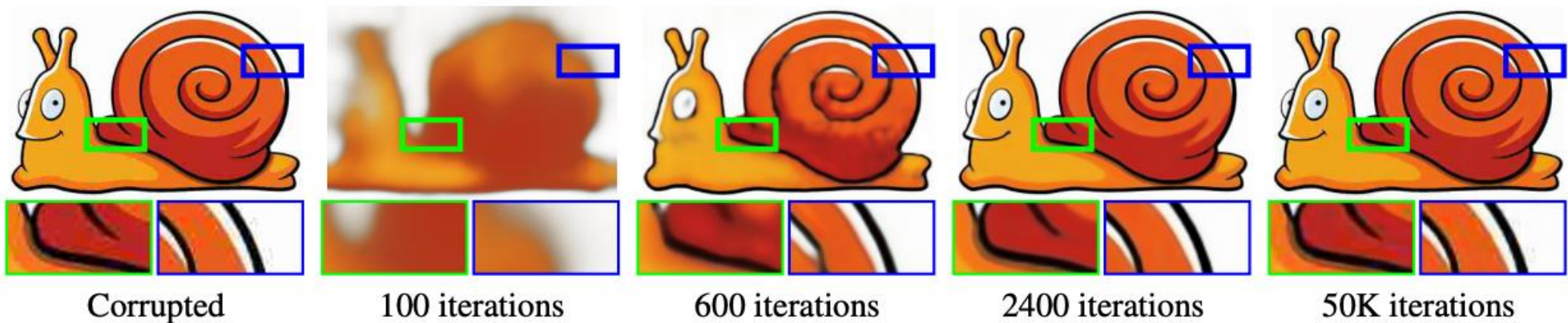
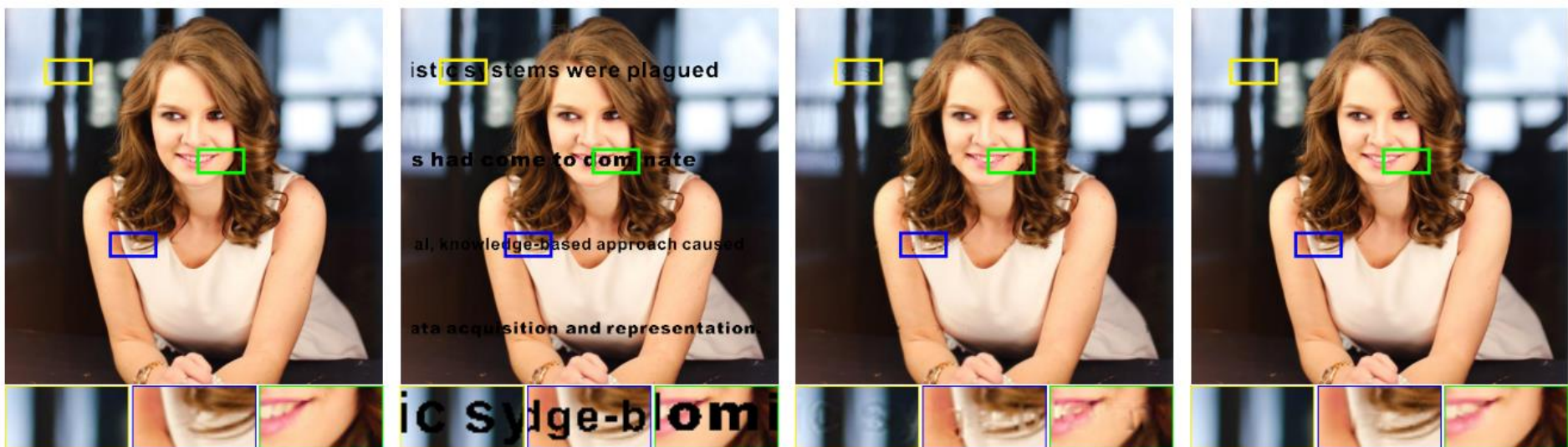


Figure 3: **Blind restoration of a JPEG-compressed image.** (*electronic zoom-in recommended*) Our approach can restore an image with a complex degradation (JPEG compression in this case). As the optimization process progresses, the deep image prior allows to recover most of the signal while getting rid of halos and blockiness (after 2400 iterations) before eventually overfitting to the input (at 50K iterations).

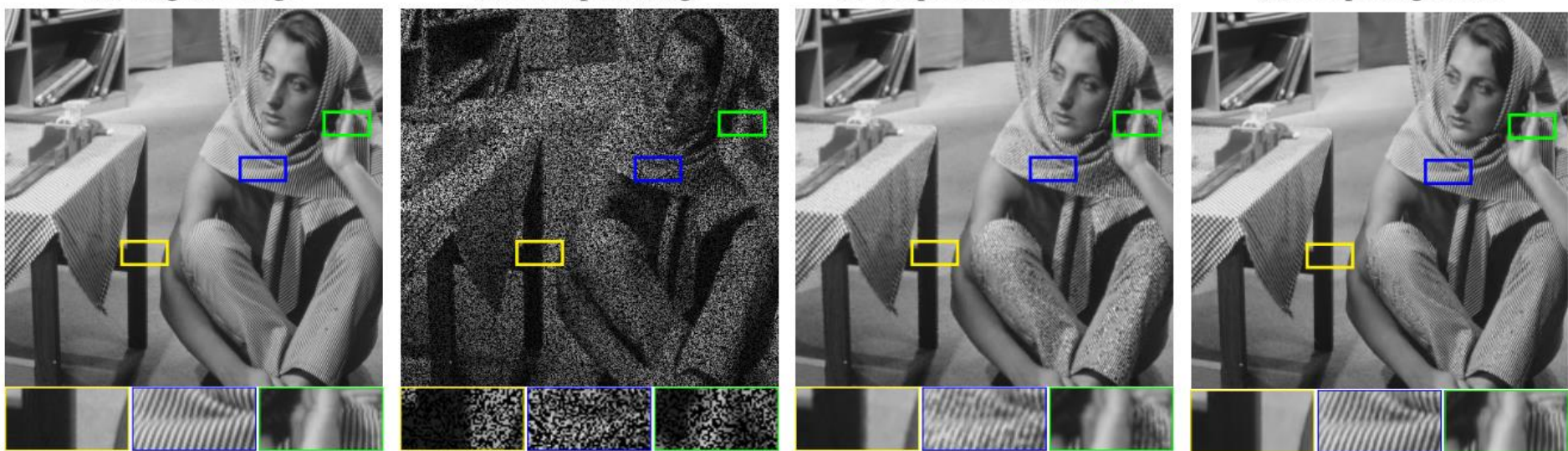


(a) Original image

(b) Corrupted image

(c) Shepard networks [27]

(d) Deep Image Prior



(e) Original image

(f) Corrupted image

(g) [25], PSNR = 28.1

(h) Deep Img. Prior, PSNR = 32.22

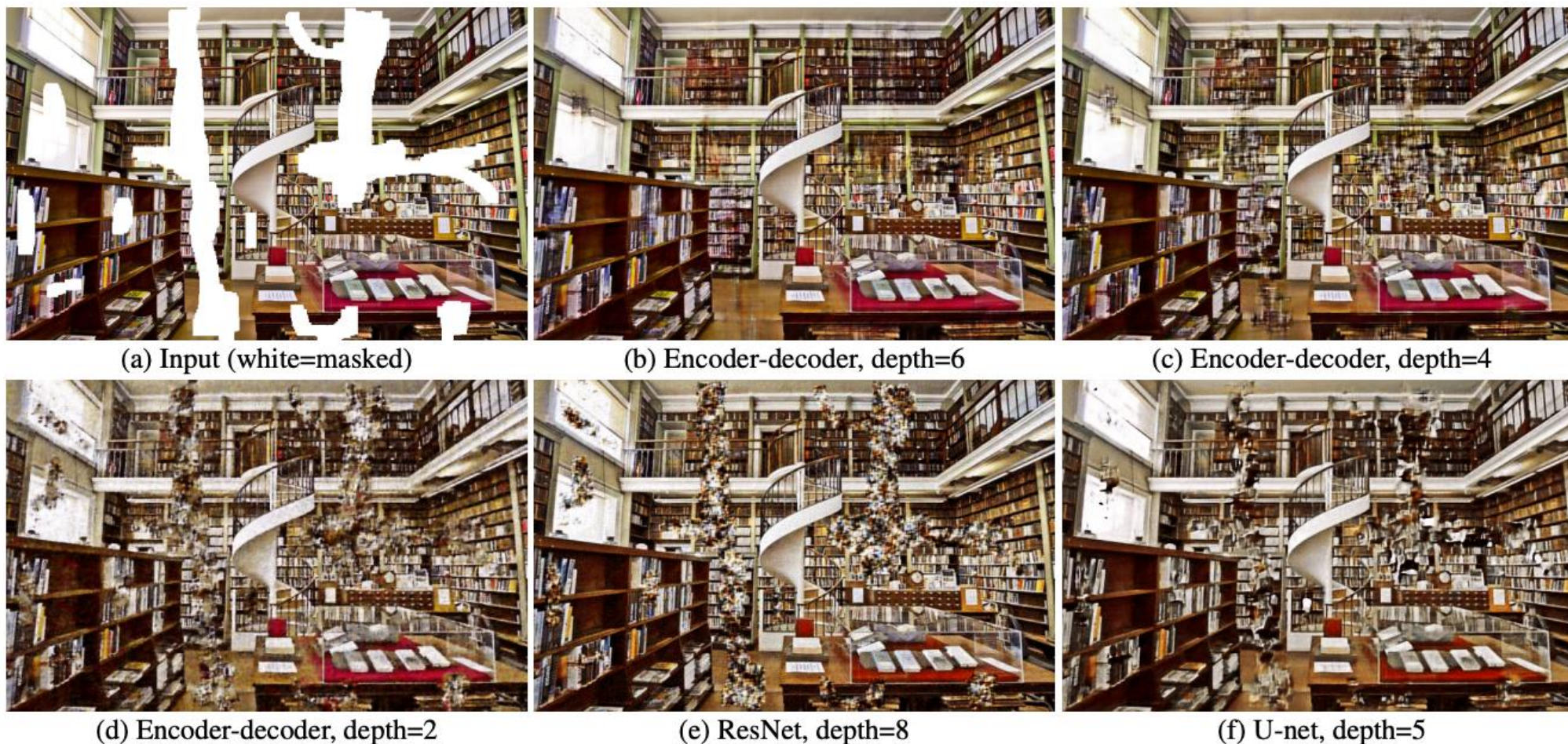


Figure 8: **Inpainting using different depths and architectures.** The figure shows that much better inpainting results can be obtained by using deeper random networks. However, adding skip connections to ResNet in U-Net is highly detrimental.

Noise2Noise: Learning Image Restoration without Clean Data

[1] Noise2Noise: Learning Image Restoration without Clean Data (ICML 2018), Lehtinen et al.

Noise2Noise: Limitation of Supervised Denoising

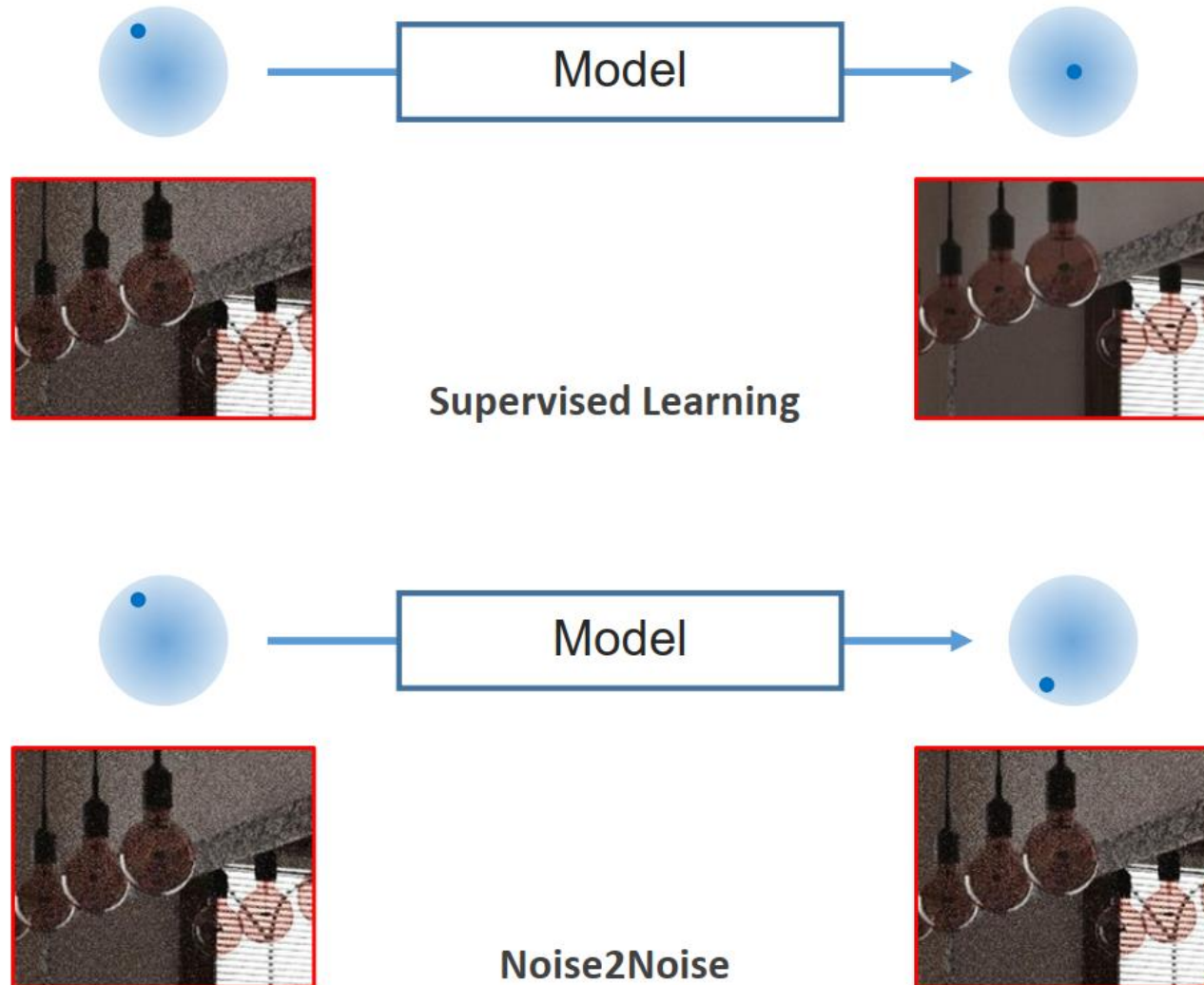
- Before Noise2Noise, DL based Denoising methods tend to solve:

$$\arg \min_{\theta} \sum \mathcal{L}(f_{\theta}(y_i), x_i)$$

where x_i denotes the clean image data while y_i denotes the observed noisy data.

- The CNNs tend to learn a projection between the noisy version and the clean version of a same image.
- However, obtaining clean training targets is often difficult or tedious.

Noise2Noise: Overview of Training Denoiser



Noise2Noise: Learning Image Restoration without Clean Data

- In Noise2Noise, the authors proposed that : “we observe that we can often learn to turn bad images into good images by only looking at bad images, and do this just as well – sometimes even better – as if we were using clean” ?

Noise2Noise: Theoretical Background

- Theoretical Background

Suppose we are estimating temperature in a room, and get a series of observations with noise $\vec{x} = \{x_1, x_2, \dots, x_n\}$, then the most common strategy is to look for a value z that satisfy:

$$\arg \min_z \mathbb{E}_x \left\{ \sum L(z, x_i) \right\}$$

- In this case, the estimation of z highly depend on the definition of loss function $L(z, \vec{x})$. For L2 norm, its optimum is the **mean** of observation of $\vec{x}$, and for L1 norm, its optimum is the **median** instead.



Noise2Noise: Generalization to Image Denoising Task

- Training neural network regressors is a generalization of this point estimation procedure:

$$\arg \min_{\theta} \mathbb{E}_{(x,y)} \{\mathcal{L}(f_{\theta}(x), y)\}$$

where x still denotes the noisy image data while y denotes the underlying clean data.

- For a sample, if we remove the details of the input data and use a simple function to represent the output of a learning, then the task goal can be set as the theoretical formula in previous page.

Noise2Noise: Generalization to Image Denoising Task

- To minimize the problem, the above formula is equivalent to the following formula:

$$\arg \min_{\theta} \mathbb{E}_x \{ \mathbb{E}_{y|x} \{ L(f_{\theta}(x), y) \} \}$$

- When using L2 loss, if $p_{y|x}$ is replaced with any distribution with the same conditional expectation, θ remains unchanged.
- So instead of using the loss between corrupted and clean image pair, the authors define a minimum average deviation from multiple measurement on a same object:

$$\arg \mathbb{E}_{x_1, x_2} \min_{\theta} L(f_{\theta}(x_1), x_2)$$

Where x_1, x_2 denote the twice observations of noisy image data on the same clean target y .

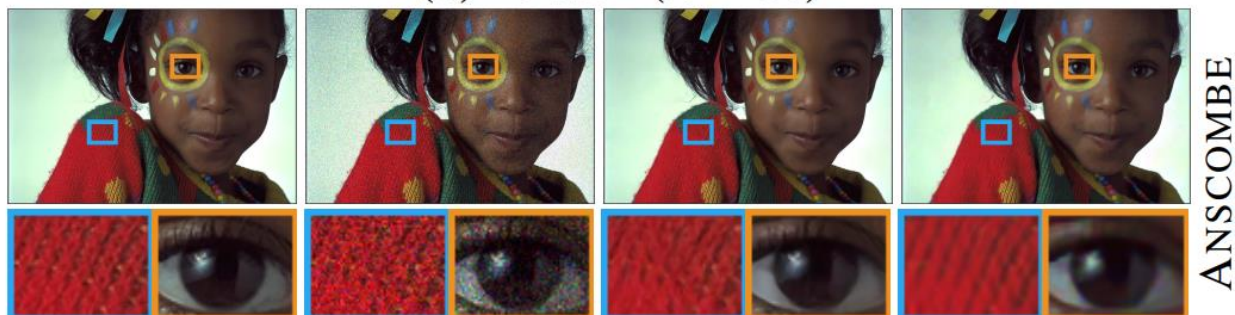


Noise2Noise: Results on simulated noisy image

(a) Gaussian ($\sigma = 25$)



(b) Poisson ($\lambda = 30$)



(c) Bernoulli ($p = 0.5$)

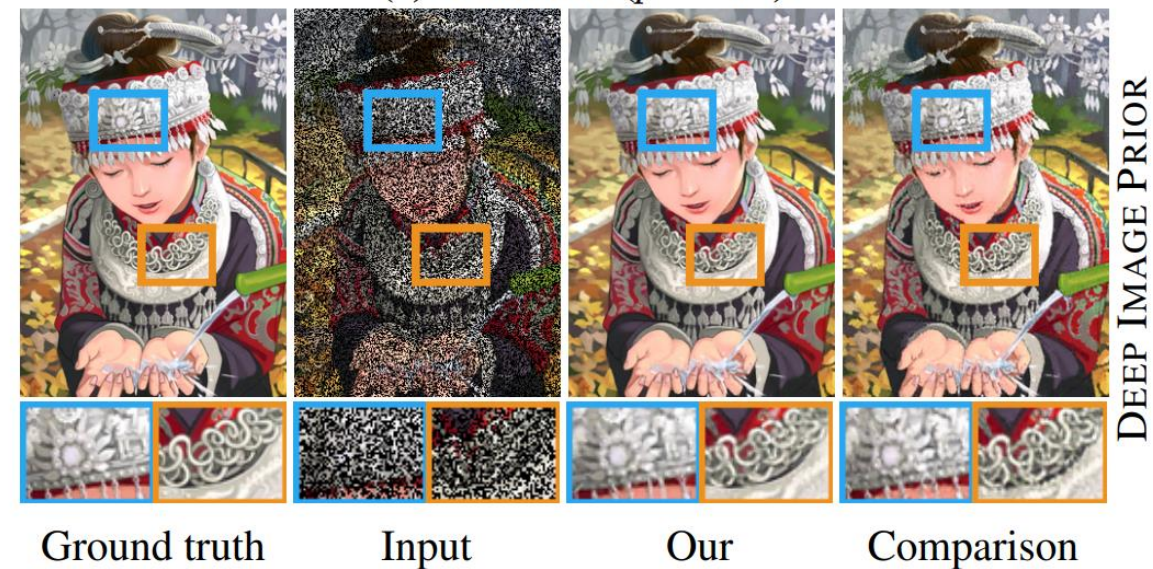


Figure 2. Example results for Gaussian, Poisson, and Bernoulli noise. Our result was computed by using noisy targets — the corresponding result with clean targets is omitted because it is virtually identical in all three cases, as discussed in the text. A different comparison method is used for each noise type.

Noise2Noise: Discussion about different loss

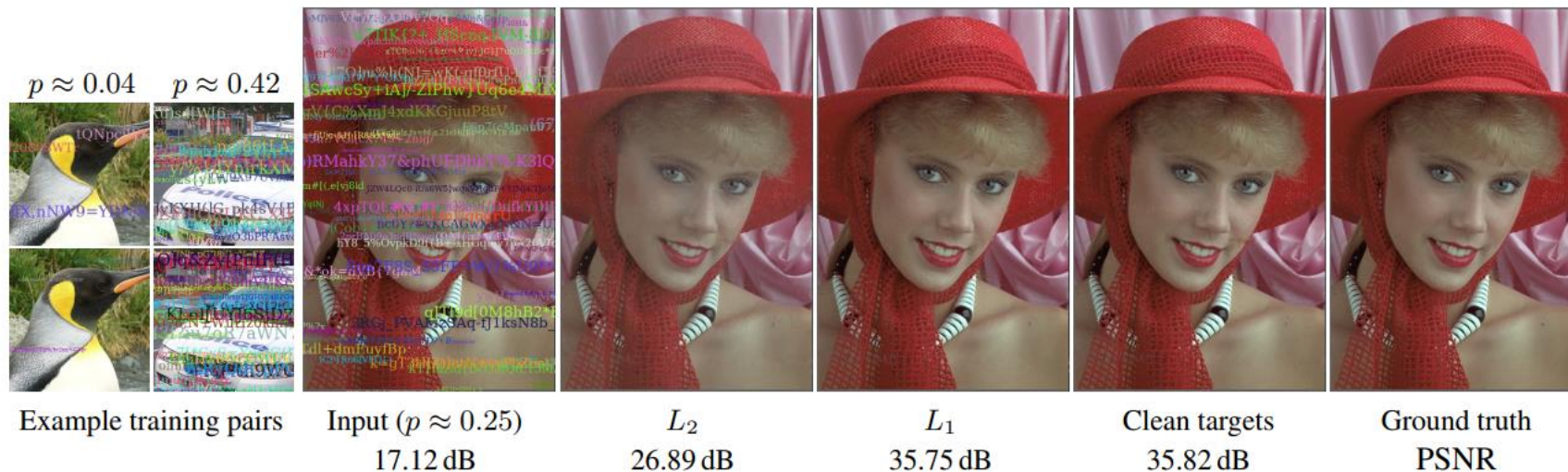


Fig. 3. Removing random text overlays corresponds to seeking the median pixel color, accomplished using the L1 loss. The mean (L2 loss) is not the correct answer.



Fig 4. For random impulse noise, the approx. mode-seeking L0 loss performs better than the mean (L2) or median (L1) seeking losses

Self-supervised Denoising Trained with Individual noisy images

- Masked-based Blind-spot Network
 - Noise2Self (ICML 2019), J Batson. et al.
 - Noise2Void (CVPR 2019), Krull A. et al.
- Novel Blind-spot Network
 - FBI-Denoiser (CVPR 2021) Byun J. et al.
 - LG-BPN (CVPR 2023) Z. Wang et al.
- Neighbor2Neighbor (CVPR 2021) T. Huang. et al.
- Noise2SR (MICCAI 2022) X. Tian et al.

Motivation of Self-supervised Denoising

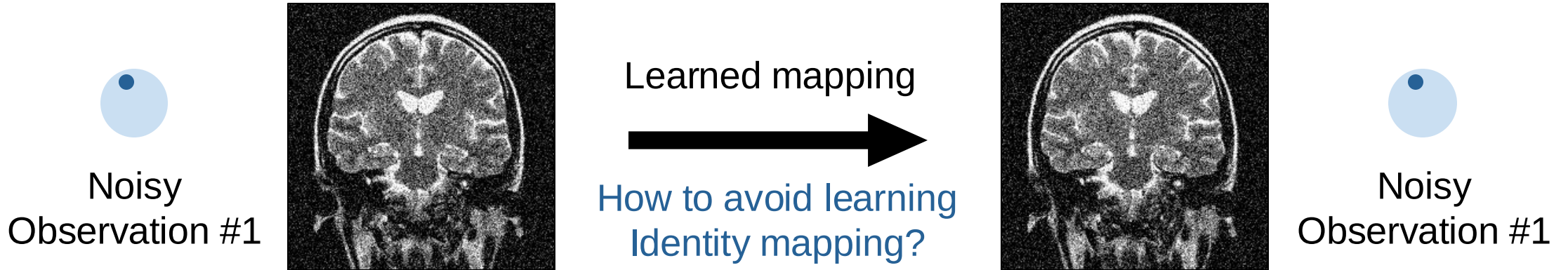
- In Biomedical Scenarios:

Acquisition of **ground-truth images** is **infeasible**

- **High electron dose** causes sample destructions
- **Long scanning time** for living subjects causes motion artifacts
- Supervised requires **High-SNR datasets**
- Noise2Noise requires **paired noisy observations datasets**

Self-supervised Denoising

- Learning denoising from an individual noisy observation



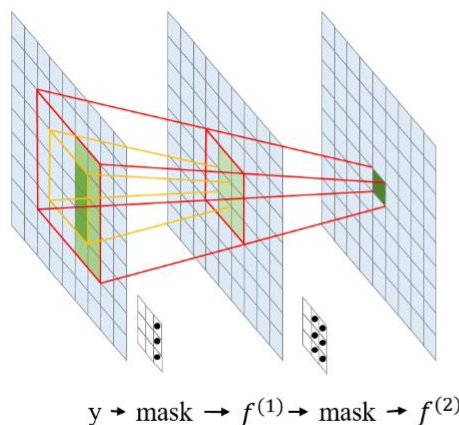
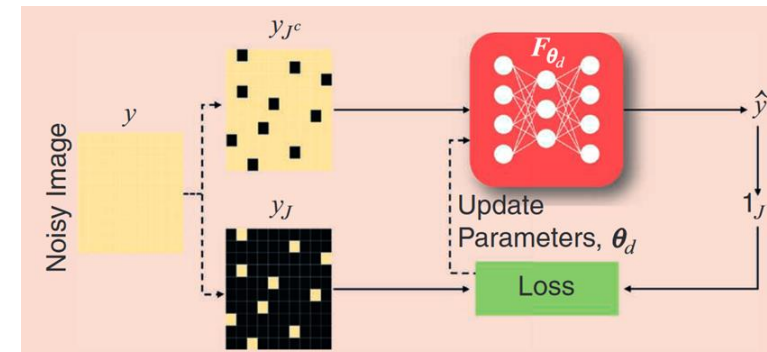
- Strategy
 - Blind-Spot Network: e.g. Noise2Void, Noise2Self, FBI-Net, etc.
 - Create paired noisy image: Neighbor2Neighbor, etc.

Assumptions of self-supervised learning

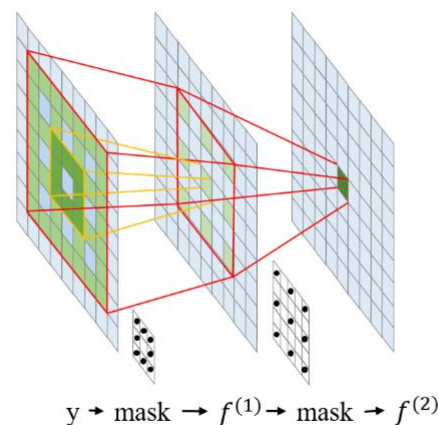
- Pixel values in $\mathbf{x}$ are not **statically independent**.
- Noise is **pixel-wise independent**
- Noise is **Zero-Mean** $\mathbb{E}[n_i] = 0 \Rightarrow \mathbb{E}[y_i] = x_i$
- Based on the assumptions that
 - Self-supervised yield a **same solution** as Supervised
 - Blind-spot network (BSN) can predict the masked pixels from its neighbor's pixels

Blind-spot Network

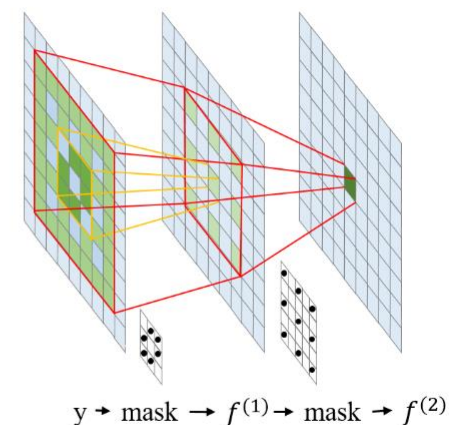
- Masked based BSN
 - E.g. Noise2Void, Noise2Self
- Novel Network Design
 - combination of centrally masked convolution & dilated convolutions



(a) Laine et al. [106]

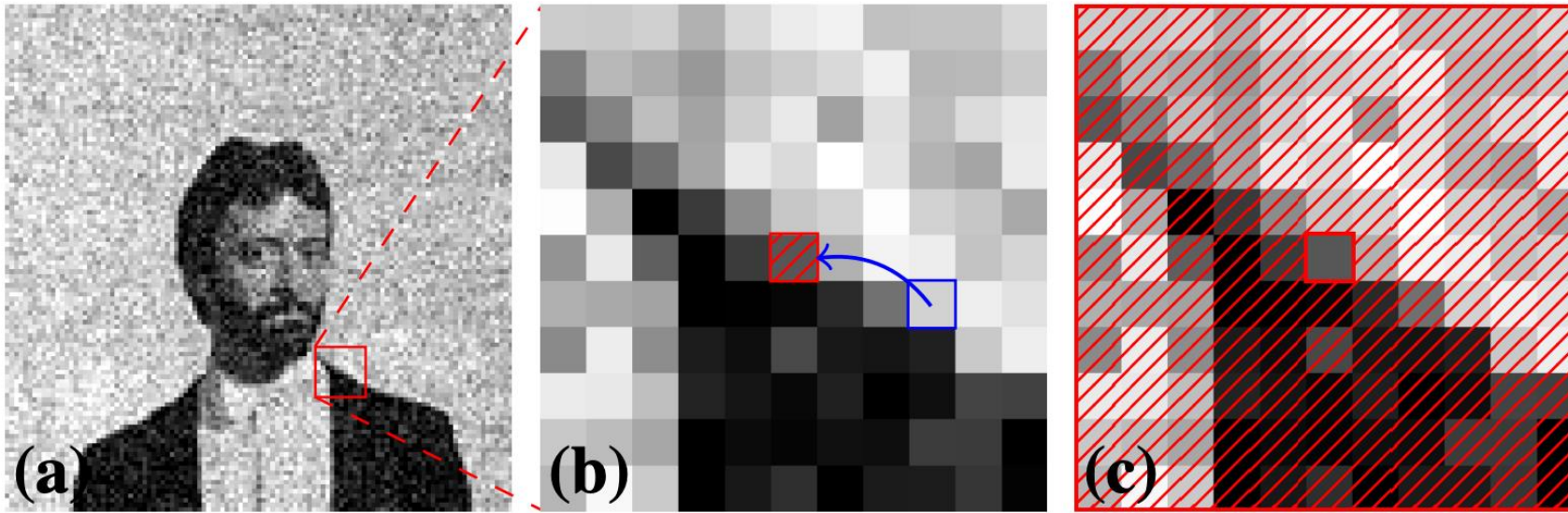


(b) DBSN[107], AP-BSN[112], LG-BSN[97], Li et al.[101]



(c) MM-BSN[98]

Noise2Void: Principle



- The network is then trained by minimizing the empirical risk:

$$\arg \min_{\theta} \sum_{n=1}^N \sum_{i=1}^{HW} L(f(X_{RF(i)}^n / x_i^n), x_i^n)$$

where $X_{RF(i)}^n / x_i^n$ denotes image patch has its center pixel x_i^n masked, and the training target is the center pixel x_i^n

	Ground Truth	Input	BM3D	Traditional	NOISE2NOISE	NOISE2VOID
BSD68						
Simulated Data						
cryo-TEM						
CTC-MSC						
CTC-N2DH						



FBI-Denoiser

- More efficient BSN architecture (FBI-Net). Reducing Inference Time.

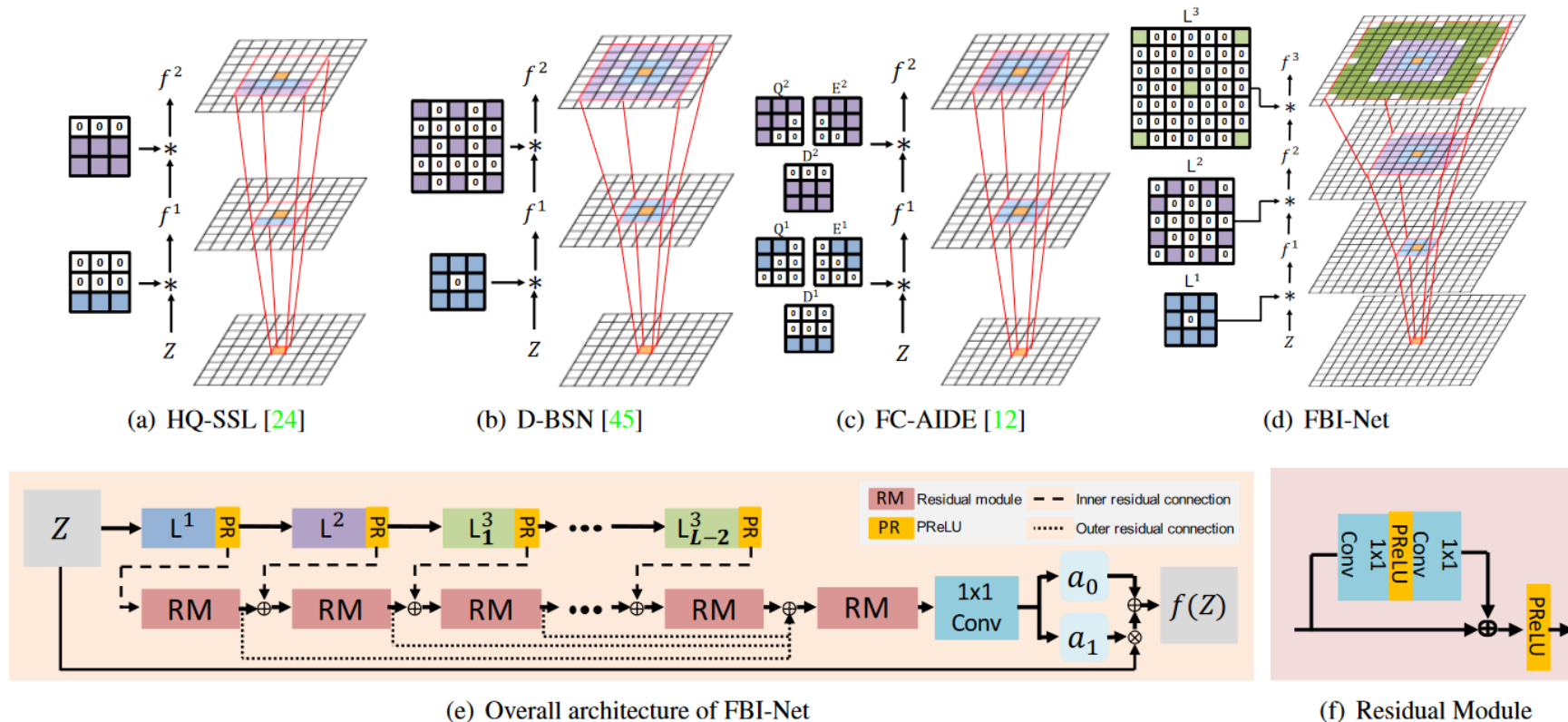


Figure 3. The layers of previously proposed blind spot network and our FBI-Net.

Dataset			GAT+BM3D	N2V	D-BSN	BP-AIDE	FBI-D	FBI-D (Sup)
Synthetic	Fivek (α, σ)	(0.01, 0.0002)	42.33 /0.9170	30.39 /0.8541	44.50 /0.9602	43.54 /0.9464	44.43 /0.9569	44.80 /0.9600
		(0.01, 0.02)	41.26 /0.8857	29.21 /0.8168	38.06 /0.8280	42.59 /0.9350	43.14 /0.9402	44.38 /0.9537
		(0.05, 0.02)	36.42 /0.8034	25.84 /0.7386	26.13 /0.4363	38.37 /0.9107	39.30 /0.9165	42.17 /0.9402
		Mixture noise	46.94 /0.9560	33.43 /0.8949	39.91 /0.8336	46.43 /0.9658	46.87 /0.9695	47.55 /0.9702
Real	FMD	CF FISH	31.31 /0.8920	31.92 /0.8831	22.86 /0.4479	32.17 /0.8841	32.22 /0.8853	32.98 /0.9111
		CF MICE	37.20 /0.9617	37.54 /0.9611	30.61 /0.7305	38.31 /0.9634	38.32 /0.9637	38.95 /0.9669
		TP MICE	33.76 /0.9157	33.34 /0.9080	26.24 /0.4239	33.89 /0.9023	33.95 /0.9084	34.40 /0.9217
Inference time			5.13s	0.06s	0.99s	2.00s	0.21s	

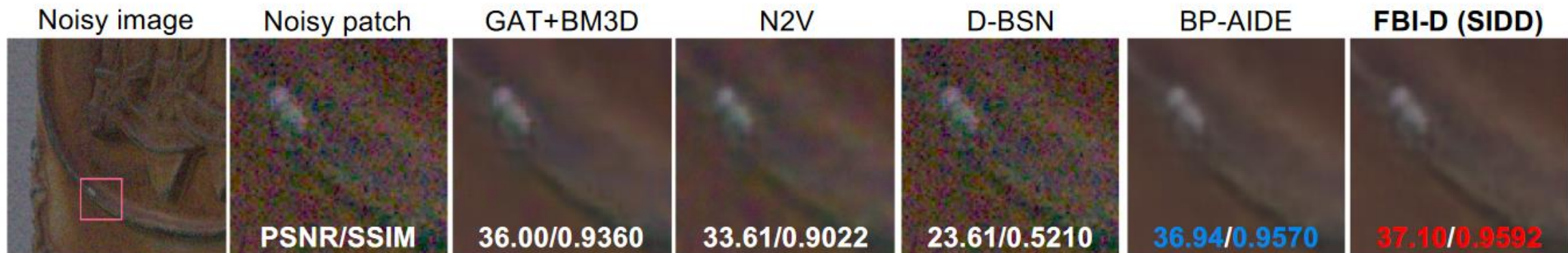
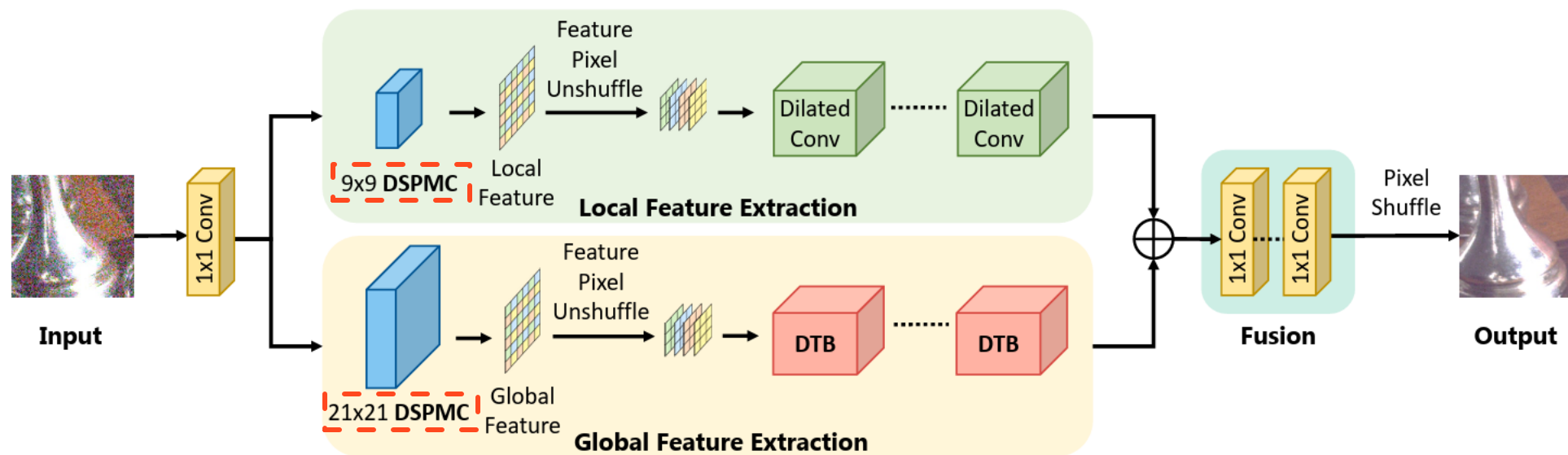


Figure 5. Visualization results of DND

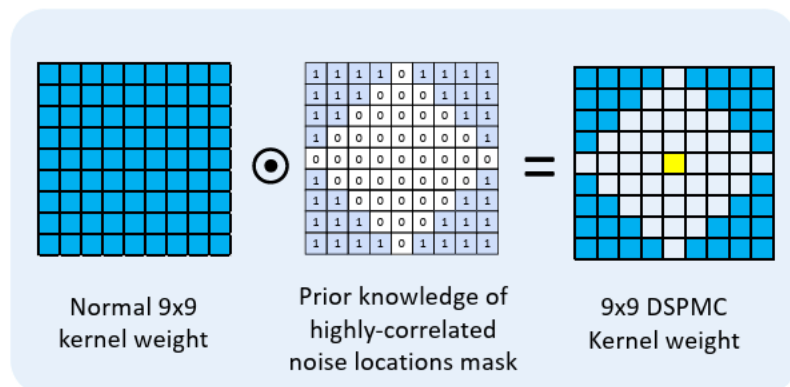
LG-BPN: Local and Global Blind-Patch Network



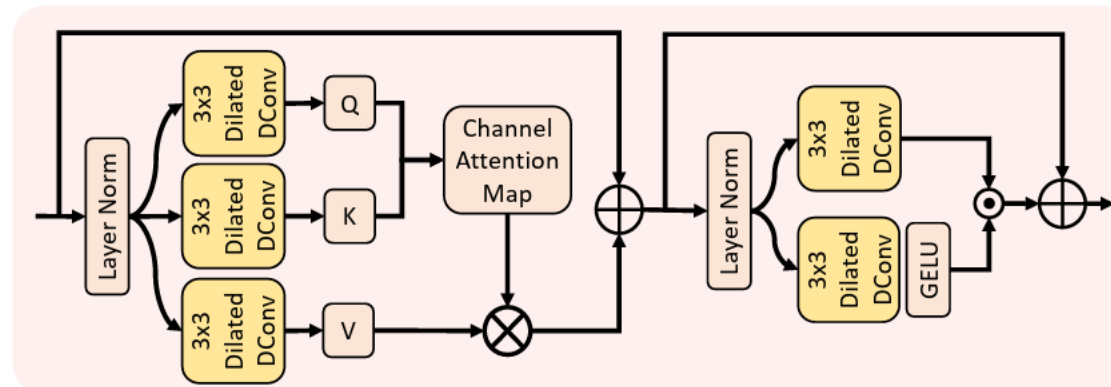
Adopt **Larger** Convolution Kernel



Provide Larger receptive field for central pixel



Densely-Sampled Patch-Masked Convolution (DSPMC)



Dilated Transformer Block (DTB)

Neighbor2Neighbor : Similarity-based Self-Learning for Image Denoising

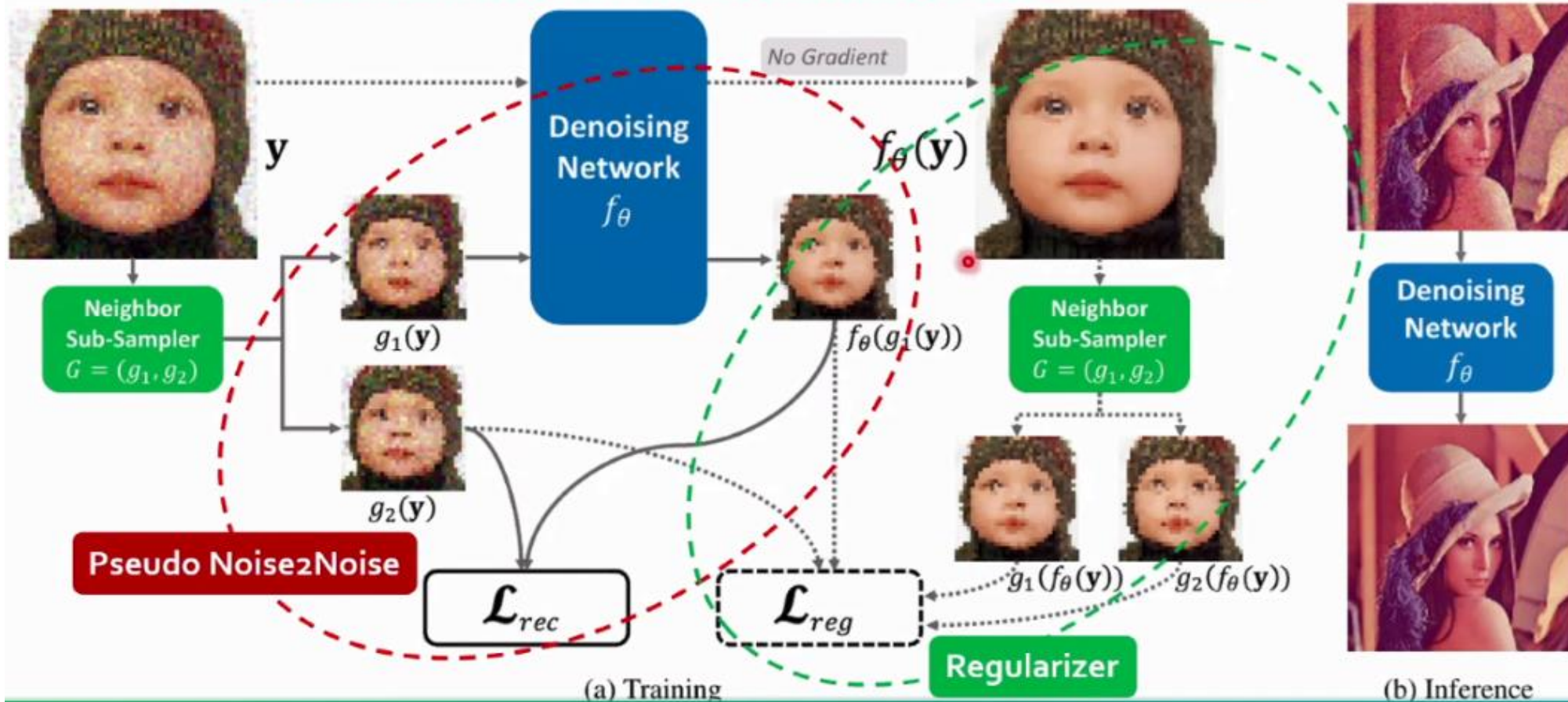
[1] Huang, Tao, et al. "Neighbor2neighbor: Self-supervised denoising from single noisy images." CVPR 2021.

Neighbor2Neighbor: Principle

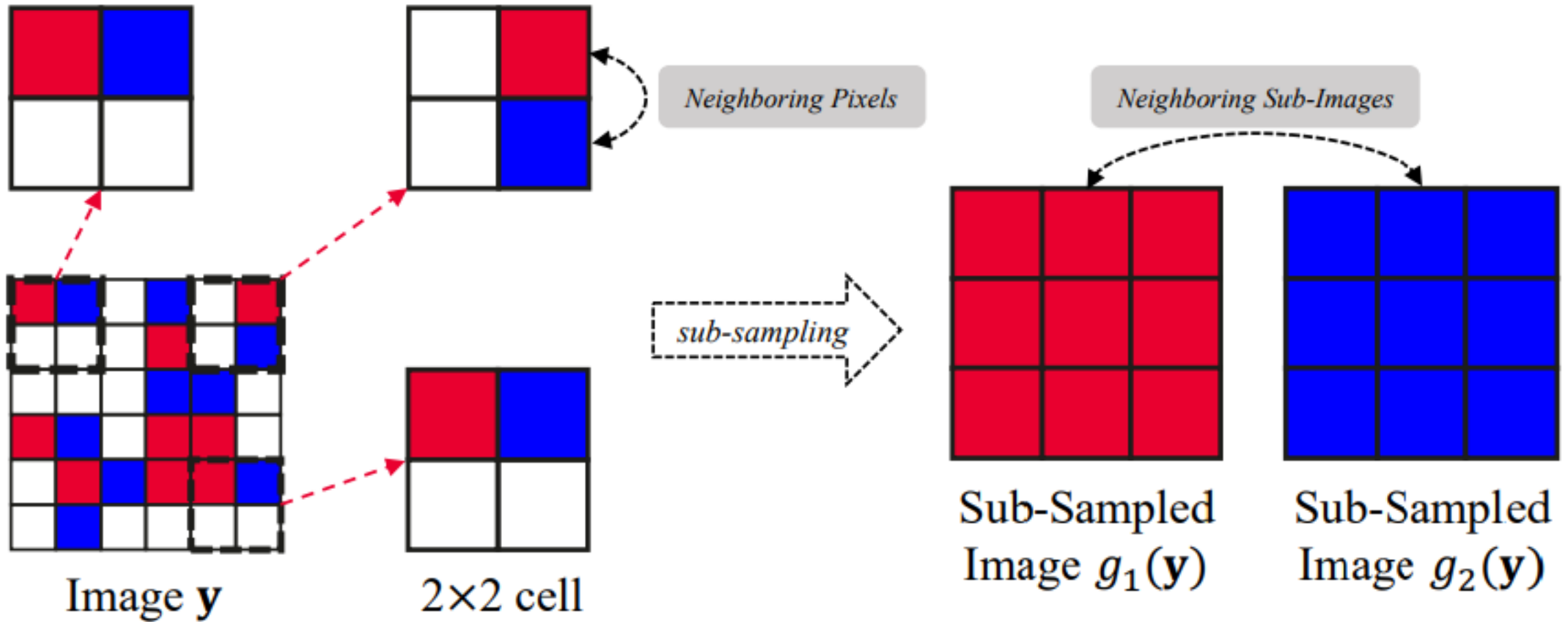
$$\mathcal{L} = \mathcal{L}_{rec} + \gamma \cdot \mathcal{L}_{reg}$$

$$= \underbrace{\|f_{\theta}(g_1(\mathbf{y})) - g_2(\mathbf{y})\|_2^2}_{\text{Pseudo Noise2Noise}}$$

$$+ \underbrace{\gamma \cdot \|f_{\theta}(g_1(\mathbf{y})) - g_2(\mathbf{y}) - (g_1(f_{\theta}(\mathbf{y})) - g_2(f_{\theta}(\mathbf{y})))\|_2^2}_{\text{Regularizer}}$$



Neighbor2Neighbor: Implementation detail



Neighbor2Neighbor: Loss function design

$$\begin{aligned} & \mathbb{E}_{\mathbf{y}|\mathbf{x}} \{f_{\theta}^*(g_1(\mathbf{y})) - g_2(\mathbf{y}) - (g_1(f_{\theta}^*(\mathbf{y})) - g_2(f_{\theta}^*(\mathbf{y})))\} \quad (4) \\ &= g_1(\mathbf{x}) - \mathbb{E}_{\mathbf{y}|\mathbf{x}}\{g_2(\mathbf{y})\} - (g_1(\mathbf{x}) - g_2(\mathbf{x})) \\ &= g_2(\mathbf{x}) - \mathbb{E}_{\mathbf{y}|\mathbf{x}}\{g_2(\mathbf{y})\} = 0. \end{aligned}$$

With the last two terms in Equation (4), we consider the gap between the ground truths of the training image pair. If the gap is zero, the subtraction of the last two terms in Equation (4) vanishes, and Equation (4) becomes a special case of Noise2Noise paired training in Equation (1). However, if the gap is non-zero, these two terms serve as a correction of the ground truth gap between the first two terms in Equation (4), forcing (4) to be zero.



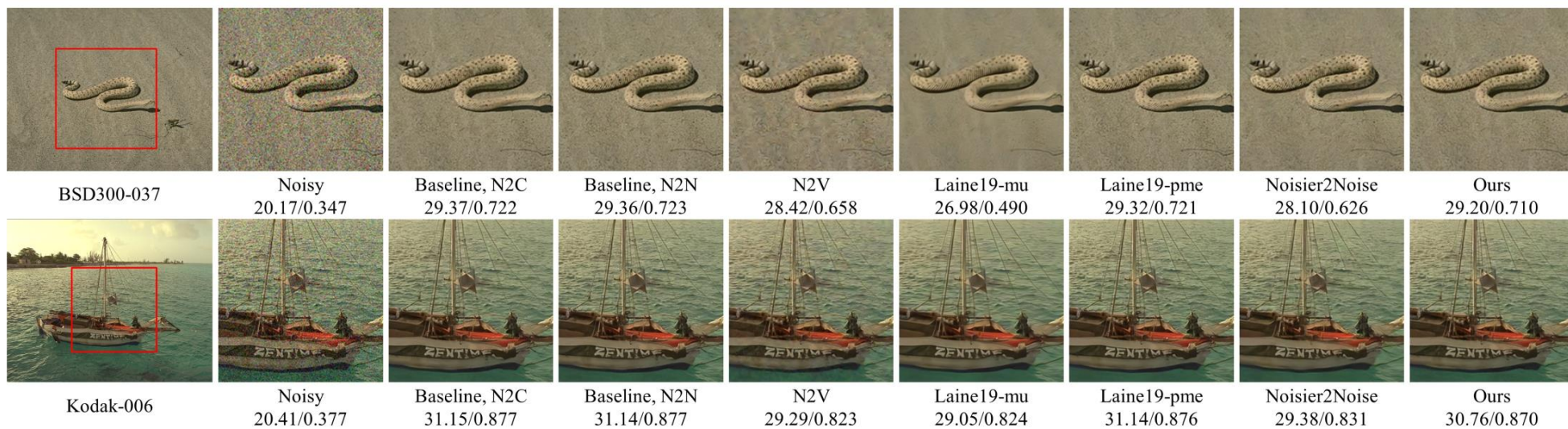


Figure 3: Visual comparison of our method against other competing methods in the setting of Gaussian $\sigma = 25$. The quantitative PSNR(dB)/SSIM results are listed underneath the images. Best viewed in color.

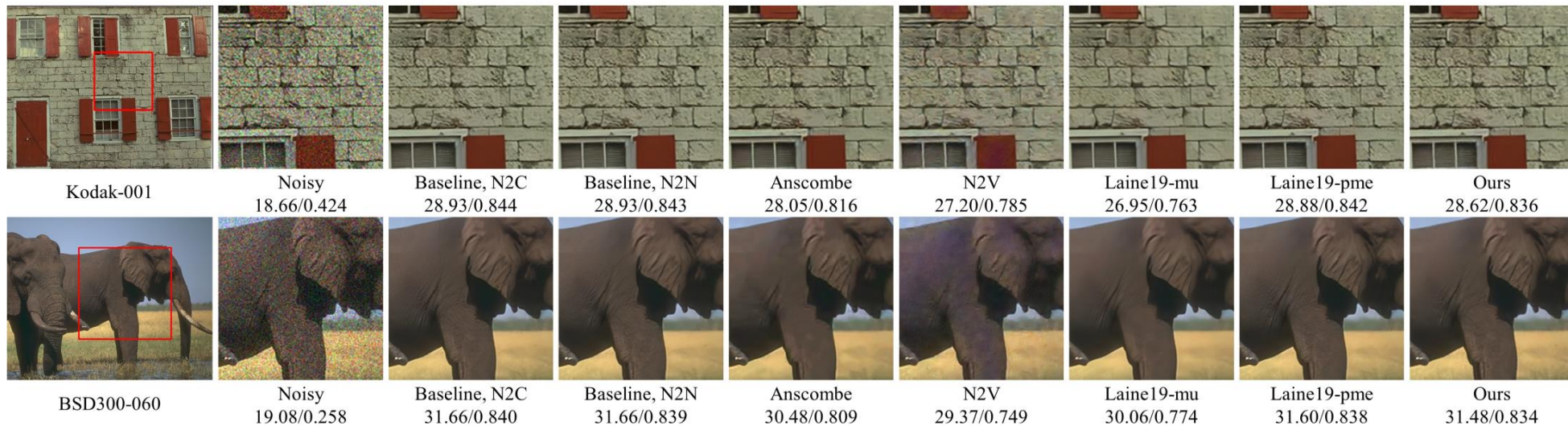


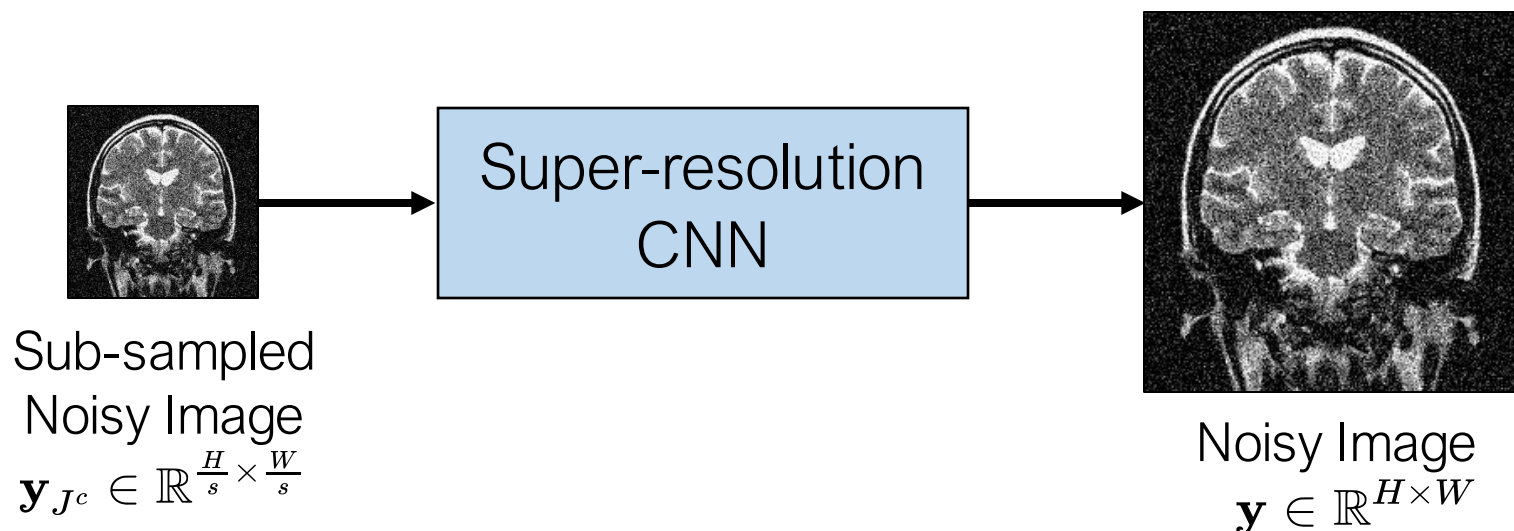
Figure 4: Visual comparison of our method against other competing methods in the setting of Poisson $\lambda = 30$. The quantitative PSNR(dB)/SSIM results are listed underneath the images. Best viewed in color.

Noise2SR: learning to denoise from super-resolved single noisy fluorescence image

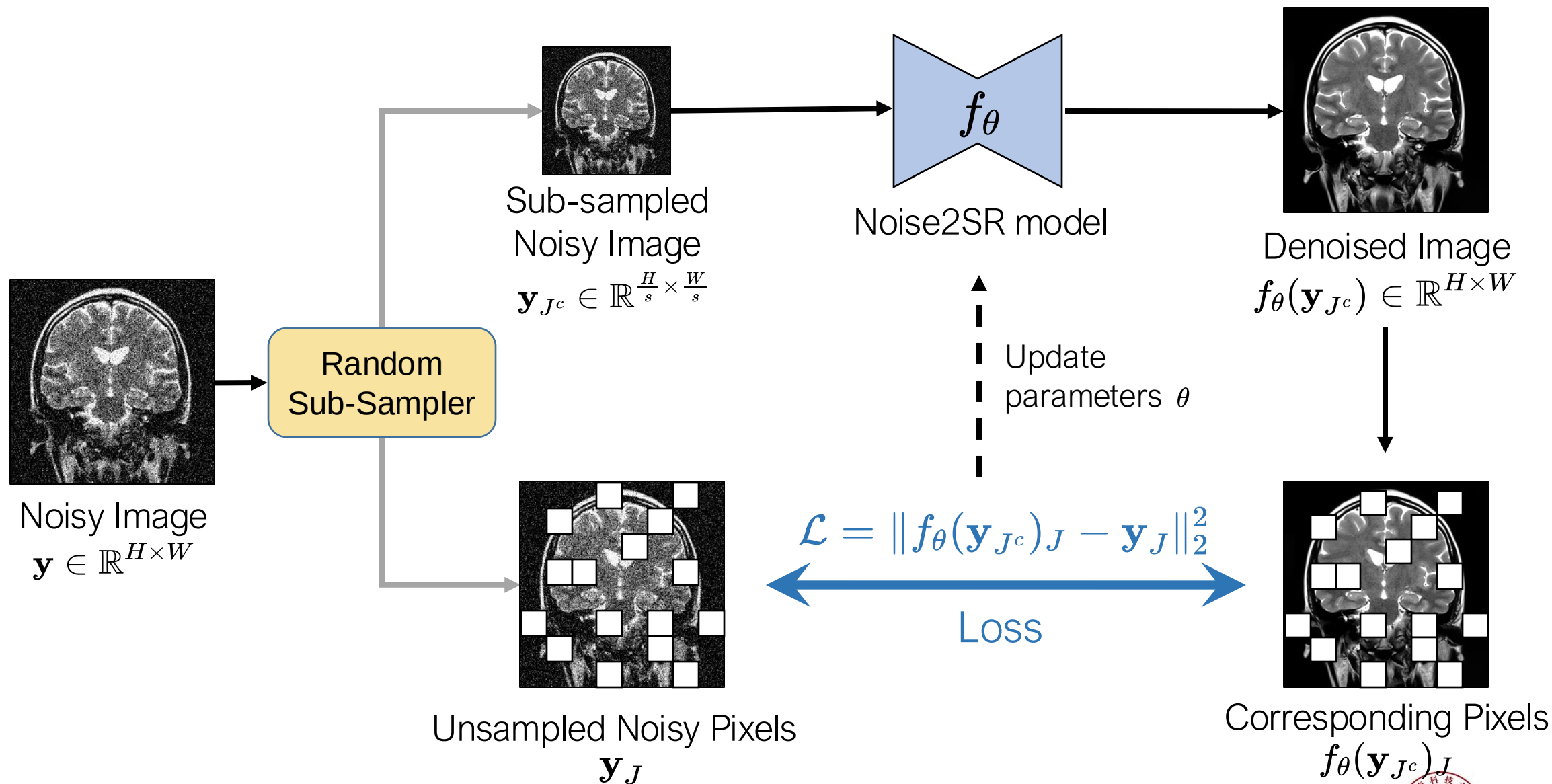
[1] Tian, Xuanyu, et al. "Noise2SR: learning to denoise from super-resolved single noisy fluorescence image." (MICCAI 2022)

Noise2SR: Principle

- Paired noisy images with **different resolutions** for training
 - Sub-sampled noisy image generated from original noisy image
- Maintain the **continuous structure** of images
- Sub-sampling **alleviate** the spatial correlation of noise
- Super-resolution Strategy: **Capture high frequency information**

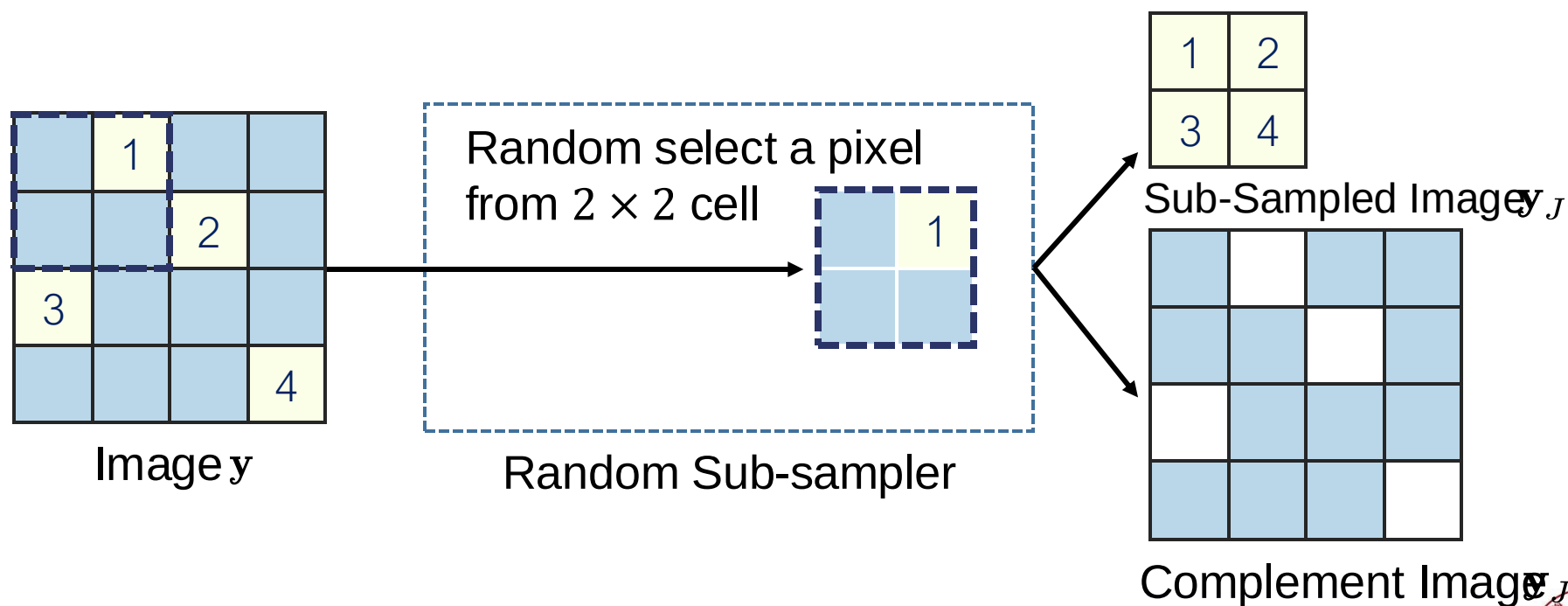


Noise2SR: Framework

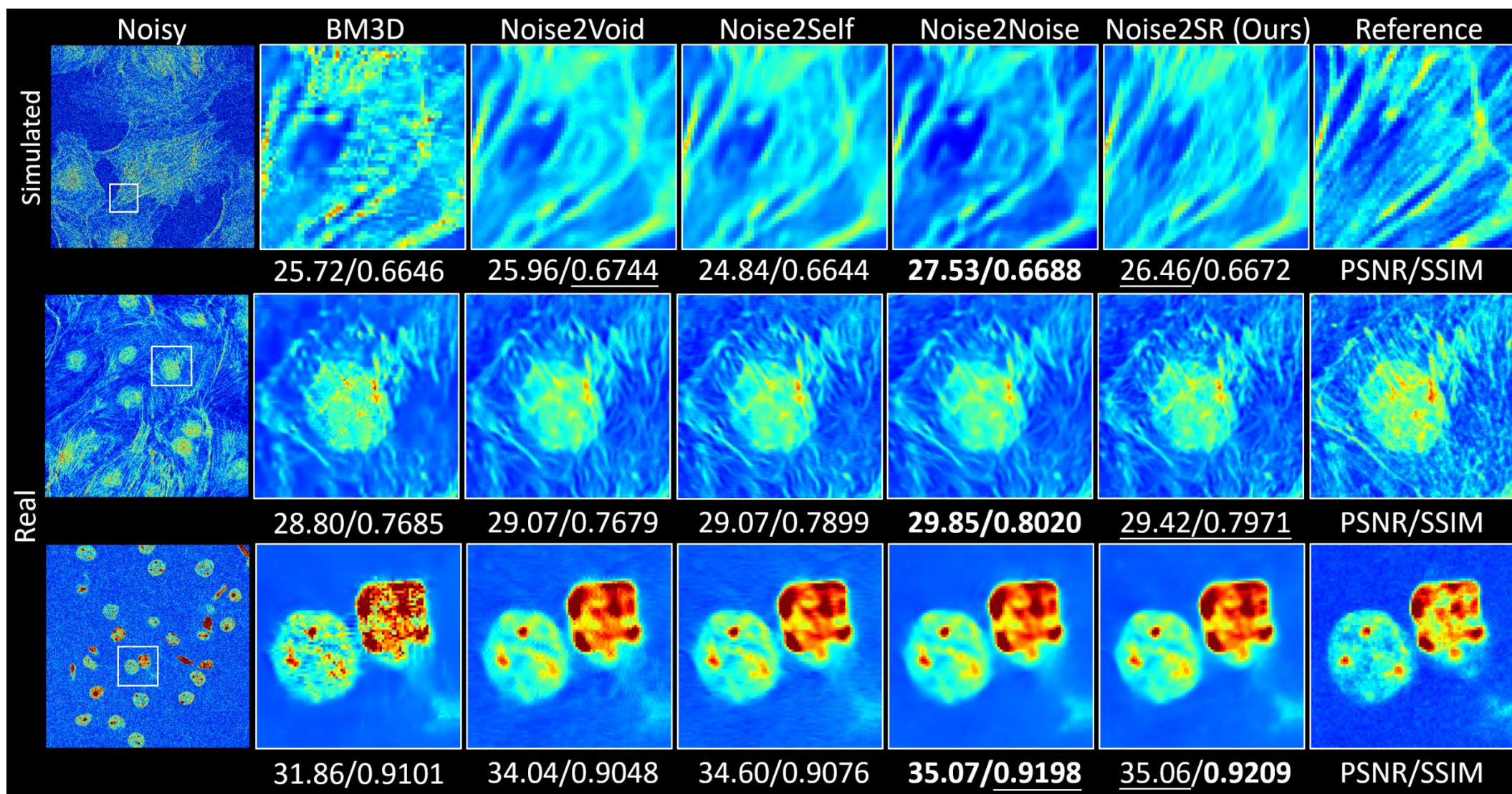


Noise2SR: Sub-sampler Module

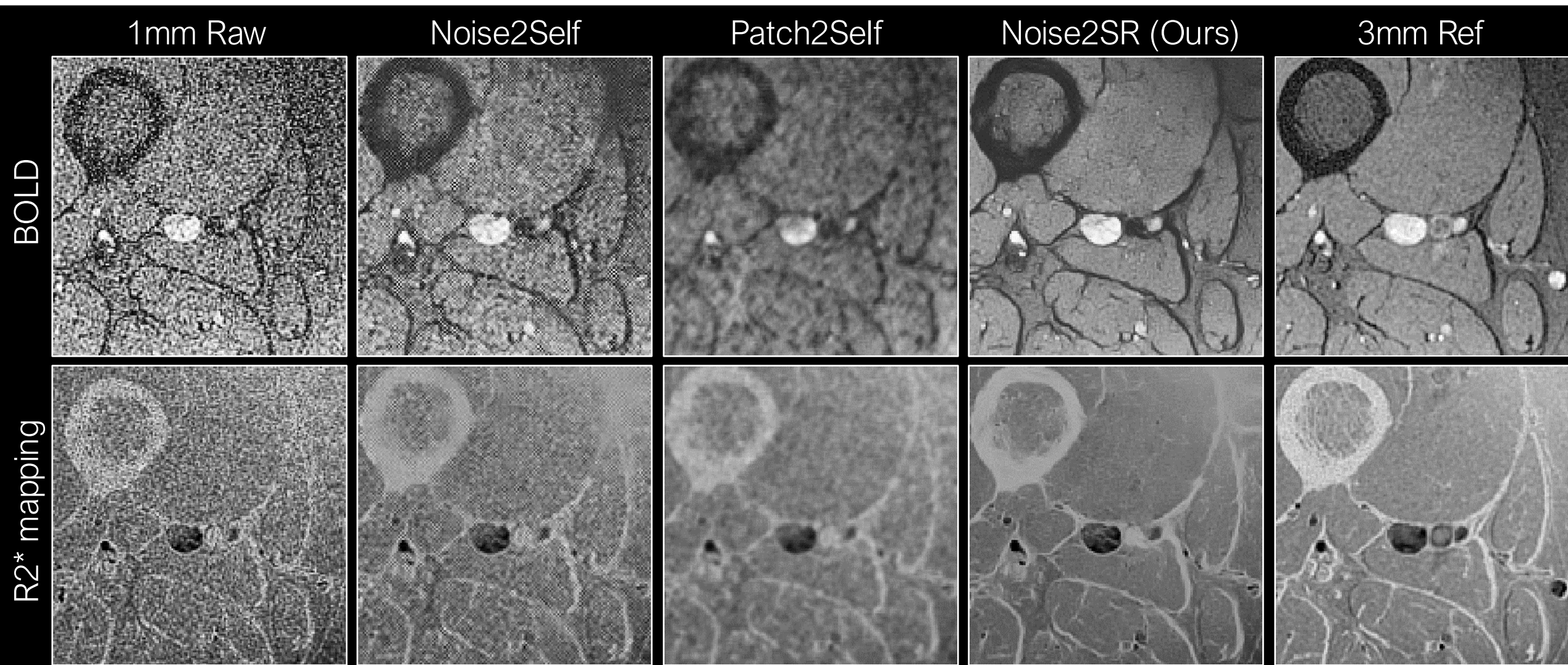
- Divide the image into multiple cells with size of 2×2 ;
- Select a pixel from a cell **randomly**;
- Generate the Sub-sampled image y_J and complement image y_J^c .



Noise2SR: Fluorescence Microscopy Denoising



Noise2SR: Results of BOLD image denoising



Take home message

- **Noise2Void** : the unsupervised Noise2Void method requires single noisy images for training and more assumptions as follows:
 - ❖ Zero-mean noise, the same as for Noise2Noise;
 - ❖ Conditional independence of noise given the signal values, i.e., $p(n|s) = \prod_i p(n_i|s_i)$;
 - ❖ Local predictability, meaning that two pixels i and j (within a certain distance) are not independent, i.e., $p(s_i|s_j) \neq p(s_i)$.
- **Noise2Sim** : the unsupervised Noise2Void method requires single noisy images for training and more assumptions as follows:
 - ❖ Existence of non-local similar image patches.